Semiparametric SDF Estimators for Pooled, Non-Traded Cash Flows

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The authors report no conflict of interest. The authors alone are responsible for the content and writing of the paper.

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Abstract

This paper introduces a unified semiparametric framework for estimating stochastic discount factors (SDF) on non-traded cash flow streams, providing a unifying objective-function perspective on the popular approaches of Driessen et al. (2012) and Korteweg and Nagel (2016). Our nonlinear least-mean-distance estimator is accompanied by a spatial asymptotic inference framework that accounts for the near-epoch dependence of overlapping private equity funds. Using Monte Carlo simulations, we demonstrate that the non-negligible finite-sample estimation error observed at short horizons can be controlled and effectively stabilized by averaging pricing errors over an extended set of compounding dates, and by aggressively aggregating cross-sectional data into seasoned vintage-year portfolios. Empirically, we analyze a comprehensive private equity dataset covering Buyout (BO) and Venture Capital (VC) funds and reveal two critical data constraints: First, including recent, unseasoned vintages severely inflates market beta estimates due to an outsized mechanical reliance on reported net asset values. When constraining the sample to heavily seasoned funds at an MSE-minimizing 15-year horizon, we estimate an attenuated market beta of 0.79 and 0.80 for equal- and value-weighted Buyout portfolios, respectively, and 1.26 and 1.05 for Venture Capital. Second, multi-factor models and abnormal performance (α) intercepts prove fundamentally fragile due to severe finite-sample noise. Consequently, we conclude that in our sample, single-factor market models are substantially more stable than multi-factor specifications, and we recommend restricting semiparametric SDF evaluation to such models until substantially more seasoned, cross-sectional private market data becomes available.

1 Introduction

Private equity has outgrown its niche, sitting today on more than $13 trillion in assets under management (McKinsey & Company, 2024), yet rigorous asset-pricing tools have not kept pace with this ascent. The empirical analysis and risk assessment of private equity and other non-traded cash flows remain fundamentally challenging due to the absence of market-based valuations and the inherent frictions of private markets (i.e., under incomplete information). Unlike public assets with trusted and tradeable valuations (in liquid secondary markets), private equity fund investments generate irregular, infrequently observed cash flows for which standard return-based asset pricing techniques are unsuitable.

To evaluate such irregular fund cash flows, the pioneering works of Driessen et al. (2012) and Korteweg and Nagel (2016) introduced highly influential stochastic discount factor (SDF) estimators that have become the gold standard in private market asset pricing. The profound impact of their methodologies is evident from widespread adoption. The Driessen et al. (2012) and Korteweg and Nagel (2016) estimators are heavily utilized to price a wide array of non-traded assets, spanning private equity, venture capital and real estate to infrastructure and collateralized

loan obligations. Building on these major contributions, we propose a unified semi-parametric framework that recovers both estimators as special cases under specific aggregation and loss-function choices. The core benefit of our unified framework lies in generalizing the set of discounting dates, which ultimately allows researchers to identify, comprehend, and control the small-sample bias inherent to cash-flow-based SDF pricing.

Our nonlinear least squares estimator stems from the class of Least-Mean-Distance (LMD) estimators, which are easier to handle than traditional Generalized Method of Moments (GMM) approaches (Pötscher and Prucha, 1997). Our LMD estimator both simplifies and extends the GMM methodology of Driessen et al. (2012), and we provide the spatial asymptotic inference framework that was missing in the original paper. The asymptotic inference formulations rely on the concept of spatial (near-epoch) dependence between funds as proposed by Korteweg and Nagel (2016). In this context, it is paramount to quantify the economic distance between funds by a measure like absolute vintage year difference or cash flow overlap[1].

For a typically sized dataset of private equity funds, we document a severe finite-sample bias in cash-flow-based SDF estimators like Driessen et al. (2012) and Korteweg and Nagel (2016) that inherently arises from the non-linearity of the estimating equations. While this bias can be mitigated by increasing the cross-sectional sample size, such data is rarely available. Instead, we demonstrate that by generalizing the estimator to average pricing errors over multiple, non-optimal discounting dates, researchers can deliberately introduce a Timing Risk Premium that effectively offsets this finite-sample bias, dramatically improving the estimates in small samples (in our simulation study).

In the empirical application of our new estimator, we evaluate simple linear and exponential affine SDF models utilizing the well-known Preqin cash flow database across a grid of compounding horizons. While evaluating multi-factor models proves unreasonable given high estimator variance, our single-factor market models yield stable and economically meaningful estimates for both Buyout and Venture Capital funds. At an MSE-minimizing 15-year horizon upon seasoned vintage years, we estimate a market beta of 0.79 and 0.80 for equal- and value-weighted Buyout portfolios, and 1.26 and 1.05 for Venture Capital, respectively. While our rigorous asymptotic inference framework yields conservative standard errors, hv-block cross-validation stability diagnostics indicate that these market beta estimates are robust across data subsets. We document that the inherent estimator variance limiting multi-factor approaches can only be partially overcome by forming portfolios with a larger number of cross-sectional units per vintage year, highlighting a fundamental data constraint in modern private market asset pricing.

The paper is structured as follows. Section 1.1 reviews the related literature. Section 2 introduces our semiparametric LMD estimator and its corresponding asymptotic inference framework. Section 3 applies the method to estimate q^5-investment-factor SDFs for private equity fund data using simulated and real-world cash flows, with Buyout and Venture Capital results presented in Section 4. Section 5 concludes. Appendix A analyzes the bias–variance tradeoff in the context of our SDF

[1]Note that this economic inter-fund *distance* is conceptually distinct from the statistical "distance" minimized by the Least-Mean-Distance estimator.

estimation problem. Appendix B compares our simulation and empirical results to their Driessen et al. (2012) counterparts. Appendix **??** replicates the empirical analysis for aggregate Private Equity funds.

1.1 Related Literature

For private equity funds, performance evaluation under the SDF framework remains an active research topic (Driessen et al., 2012; Korteweg and Nagel, 2016, 2024; Giommetti and Jørgensen, 2021; Gupta and Nieuwerburgh, 2021; Boyer et al., 2023; Korteweg et al., 2024) alongside an expanding list of applications across other private asset classes. Specifically, the Driessen et al. (2012) estimator is heavily utilized to price human capital contracts, venture capital, real estate and private equity funds (Kroencke et al., 2013; Kim et al., 2015; Farrelly and Stevenson, 2019; Lee et al., 2019; Zhao, 2020; Tausch and Pietz, 2024), while the Korteweg and Nagel (2016) approach has found broad application spanning private equity, infrastructure, real estate investment trusts, collateralized loan obligations, impact funds and private debt (Giommetti and Jørgensen, 2021; Andonov et al., 2021; Riddiough and Wiley, 2022; Boyer et al., 2023; Cordell et al., 2023; Erel et al., 2024; Jeffers et al., 2024; Korteweg and Nagel, 2024). Previously, SDF performance evaluation was already applied for public stock markets by, e.g., Farnsworth et al. (2002) and for hedge fund returns by Li et al. (2016).

Within the private equity literature, SDF-based approaches can be broadly classified into parametric and semiparametric methods. Parametric models, such as those proposed by Franzoni et al. (2012) and Ang et al. (2018), explicitly specify the return dynamics and the underlying data generating process of private equity investments. While these methods can uncover latent factors or deal-level dynamics, they typically rely on strong distributional assumptions (e.g., log-normality) that may not fully capture the irregular and skewed nature of PE cash flows.

Consequently, semiparametric approaches have gained prominence, starting with the seminal cross-sectional GMM framework of Driessen et al. (2012). They estimate a linear factor model that minimizes the squared pricing errors of PE vintage year portfolios, explicitly evaluating cash flows at fund inception. Building on this, Korteweg and Nagel (2016) introduced the Generalized Public Market Equivalent (GPME) framework, shifting the estimation of the SDF to public market data while evaluating PE cash flows in a second step. Importantly, Korteweg and Nagel (2016) were the first to apply a spatial heteroskedasticity and autocorrelation consistent (SHAC) covariance matrix estimator (Conley, 1999) to account for the economic distance (cash flow overlap) between PE funds, moving beyond simple cross-sectional bootstrapping.

Further complementing this cash flow dimension, a strand of literature models the informational frictions and smoothing inherent in reported Net Asset Values (NAVs) (Brown et al., 2019; Jenkinson et al., 2020; Brown et al., 2021). Additionally, Hüther et al. (2022) leverage a similar generalized framework to evaluate private firms using credit market equivalents. Finally, for comprehensive methodological and empirical overviews of risk and return assessment in private equity, we refer to Korteweg (2019, 2023), who also discuss the popular Dimson (1979) regression that

employs lagged returns to correct for stale valuations.

Our paper's unified LMD framework contributes to this semiparametric literature by providing a unifying loss-function perspective on both the Driessen et al. (2012) and Korteweg and Nagel (2016) approaches while supplying the rigorous increasing-domain asymptotic inference missing in earlier work (drawing on the spatial theory of Pötscher and Prucha (1997), Kim and Sun (2011), and Jenish and Prucha (2012)). While Gredil et al. (2020) highlighted that NPV-based measures are biased relative to per-period returns, we explicitly formalize this discrepancy as a "Timing Risk Premium", a concept linked to recent findings on discount-rate risk in secondary markets (Boyer et al., 2023).

2 Methodology

We now formalize the semiparametric estimator that synthesizes the approaches of Driessen et al. (2012) and Korteweg and Nagel (2016). A detailed exposition of how our unified framework departs from and improves upon these base models is provided in Section 2.5.

In a nutshell, the Driessen et al. (2012) estimator determines the estimated alpha and beta parameters of a linear factor model by minimizing the squared net present value (NPV) of all n fund cash flows in a least-squares optimization problem.

$$\left(\hat{\alpha}, \hat{\beta}\right) = \arg\min_{\alpha,\beta} \sum_{i=1}^{n} \left[NPV_i(\alpha, \beta)\right]^2 \tag{1}$$

with NPV for the ith fund

$$NPV_i(\alpha, \beta) = \sum_{t \in \{t_{0,i}, t_{0,i}+1, \dots\}} \frac{\text{CashFlow}_{i,t}}{\prod_{s=t_{0,i}+1}^{t} \left(1 + r_{f,s} + \alpha + \beta r_{\text{market},s}\right)}$$

Here, the Driessen et al. (2012) approach relies on the identifying assumption that the model is correctly specified, so that $\mathbb{E}[NPV_i(\theta_0)] = 0$.

In this Methodology section, we will show:

1. Why this estimator yields conditionally expected pricing errors of zero under a Net Present Value framework, and we analytically characterize the bias term (timing risk premium) introduced by a Net Future Value approach (Section 2.1).
2. How to modify the estimator to be able to control the bias term (Section 2.2).
3. How we can estimate asymptotic standard errors for the parameters alpha and beta (Sections 2.3 and 2.4).
4. How our estimator compares to similar approaches (Section 2.5).

2.1 Asset Pricing for Pooled Cash Flows

Let private equity fund $i = 1, 2, \dots, n$ be characterized by its net cash flows $CF_{i,t}$ (i.e., distributions minus contributions) and its net asset values $NAV_{i,t}$ with discrete time index $t = 0, 1, 2, \dots, T$. To increase the mathematical tractability of the

Symbol	Description
n	Number of cross-sectional units (funds or portfolios)
V	Number of distinct vintage years
n_v	Number of portfolios in vintage year v
v_i	Vintage year of fund/portfolio i
J	Number of deals per fund
$K_{i,j}$	Number of divestment cash flows for deal j of fund i
$CF_{i,t}$	Net cash flow of fund i at time t
Ψ_t , M_t	Multi-period and single-period SDF
$\Psi_{\tau,t}(\theta)$	SDF ratio Ψ_t/Ψ_τ, parameterized by θ
$\theta = (\alpha, \beta)$	SDF parameter vector (intercept and factor loadings)
θ_0	True parameter vector
$\delta_{i,j}$	Deal-level pricing error
$\epsilon_{\tau,i}$	Fund pricing error at discounting date τ
$\bar{\epsilon}_i$	Weighted average pricing error of fund i
a_i	Fund-level alpha (abnormal return; distinct from SDF α)
$\mathcal{T}_i$	Set of discounting dates for fund i
w_i	Weighting factor for fund i
$L(\cdot)$	Loss function (e.g., $L(x) = x^2$)
$Q_n(\theta)$	Sample objective function
$d_{i,j}$	Economic distance between units i and j
D $(= b_n)$	SHAC bandwidth parameter
H	Hessian of population objective at θ_0
Λ	Long-run covariance matrix of the score
Σ	Asymptotic covariance matrix of $\hat{\theta}$

Table 1: Key notation for the Methodology section.

problem, we assume a deal-by-deal data generating process (DGP) for CF where each fund deal consists exactly of one investment and one divestment cash flow in combination with a simple return model for the multi-period deal returns. This means the fund-level cash flow process $(CF_{i,t})_{t=0,1,\ldots,T}$ is an aggregation of deal-level cash flow pairs consisting of one negative at deal inception and at least one positive cash flow later $CF_{i,t} = \sum_j^J cf_{i,j,t}$.

Assumption 1. *Deal-level data generation process:*
1. Each fund i consists of J underlying deals.
2. Each deal is characterized by exactly one, negative investment cash flow, denoted by $\mathrm{Inv}_{i,j}$, which occurs at time $t_{i,j}^{\mathrm{Inv}} \in \{0, 1, \ldots, T-1\}$. It holds $\mathrm{Inv}_{i,j} < 0$.
3. Each deal is characterized by a positive divestment cash flow stream, denoted by $(\mathrm{Div}_{i,j,k})_{k=1,\ldots,K_{i,j}}$, which occur after the investment cash flow $t_{i,j,k}^{\mathrm{Div}} > t_{i,j}^{\mathrm{Inv}}$ for all k. It holds $\mathrm{Div}_{i,j,k} > 0$.
4. The cumulative fund cash flows are generated by summarizing over all deal-level cash flows, i.e., $\sum_{t=0}^T CF_{i,t} = \sum_{j=1}^J \left(\mathrm{Inv}_{i,j} + \sum_{k=1}^{K_{i,j}} \mathrm{Div}_{i,j,k}\right)$ for all i.

From asset pricing theory, we know that we can use a stochastic discount factor

Ψ_t to price each underlying deal

$$\mathbb{E}\left[\delta_{i,j} \,\middle|\, \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right] = 0 \quad \forall\, i, j \tag{2}$$

where we denote the deal-level pricing error by

$$\delta_{i,j} := \text{Inv}_{i,j} + \sum_{k=1}^{K_{i,j}} \frac{\Psi_{t_{i,j,k}^{\text{Div}}}}{\Psi_{t_{i,j}^{\text{Inv}}}} \text{Div}_{i,j,k} \tag{3}$$

Before we can derive the expected value for the fund-level pricing errors, let us analyze the deal-level counterparts.

Case A: Discounting deals to a date at or before investment ($\tau \leq t_{i,j}^{\text{Inv}}$) First we apply the Law of Iterated Expectations (the so-called "tower property") to determine the fair value of individual deal cash flows at time $\tau \leq t_{i,j}^{\text{Inv}}$.[2]

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right] \mid \mathcal{F}_\tau\right] \tag{4}$$

Since $\tau \leq t_{i,j}^{\text{Inv}}$, it holds that $\mathcal{F}_\tau \subseteq \mathcal{F}_{t_{i,j}^{\text{Inv}}}$. In addition, the term $\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}$ is fully known at time $t_{i,j}^{\text{Inv}}$, so it can be pulled out of this inner expectation to receive

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\underbrace{\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right]}_{=0 \text{ by Eq. 2}} \mid \mathcal{F}_\tau\right] = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \cdot 0 \mid \mathcal{F}_\tau\right] = 0 \tag{5}$$

Case B: Discounting deals to a date after investment ($\tau > t_{i,j}^{\text{Inv}}$) When the discounting/compounding date τ is after the investment date, the term $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau$ is not known at $t_{i,j}^{\text{Inv}}$ (it depends on future SDF realizations). Thus, the previous logic does not hold. Instead, we have a non-zero term that represents the accumulated realized pricing error.

Let $\tau > t_{i,j}^{\text{Inv}}$. The expectation becomes:

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right] = \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_\tau\right] \tag{6}$$

Since we condition on $\mathcal{F}_\tau$ and $t_{i,j}^{\text{Inv}} < \tau$, the SDF ratio is known and can be pulled out. Then the term $\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_\tau\right]$ is the conditional expectation of the pricing error given information available at time τ. It decomposes as:

[2] At the boundary $\tau = t_{i,j}^{\text{Inv}}$, the SDF ratio $\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau = 1$ is a trivial constant, so the covariance with $\delta_{i,j}$ is zero and the same argument applies.

$$\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_\tau\right] = \underbrace{\mathrm{Inv}_{i,j} + \sum_{t^{\mathrm{Div}}_{i,j,k} \leq \tau} \frac{\Psi_{t^{\mathrm{Div}}_{i,j,k}}}{\Psi_{t^{\mathrm{Inv}}_{i,j}}} \mathrm{Div}_{i,j,k}}_{\text{Realized Cash Flows}} + \underbrace{\mathbb{E}\left[\sum_{t^{\mathrm{Div}}_{i,j,k} > \tau} \frac{\Psi_{t^{\mathrm{Div}}_{i,j,k}}}{\Psi_{t^{\mathrm{Inv}}_{i,j}}} \mathrm{Div}_{i,j,k} \mid \mathcal{F}_\tau\right]}_{\text{Expected Future Cash Flows}} \tag{7}$$

Generally, this sum can become (is usually) non-zero because it includes realized shocks (abnormal returns) that occurred between $t^{\mathrm{Inv}}_{i,j}$ and τ.

Case C: Compounding deals to a future date Let us analyze the following Net Future Value (NFV) case, where $t^{\mathrm{Future}}_{i,j}$ denotes a compounding target date after the deal investment date (i.e., $\tau < t^{\mathrm{Inv}}_{i,j} < t^{\mathrm{Future}}_{i,j}$).

$$\begin{aligned}
\mathbb{E}\left[\frac{\Psi_{t^{\mathrm{Future}}_{i,j}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right] &= \mathbb{E}\left[\frac{\Psi_{t^{\mathrm{Inv}}_{i,j}}}{\Psi_\tau}\frac{\Psi_{t^{\mathrm{Future}}_{i,j}}}{\Psi_{t^{\mathrm{Inv}}_{i,j}}}\delta_{i,j} \mid \mathcal{F}_\tau\right] \\
&= \mathbb{E}\left[\frac{\Psi_{t^{\mathrm{Inv}}_{i,j}}}{\Psi_\tau}\mathbb{E}\left[\frac{\Psi_{t^{\mathrm{Future}}_{i,j}}}{\Psi_{t^{\mathrm{Inv}}_{i,j}}}\delta_{i,j} \mid \mathcal{F}_{t^{\mathrm{Inv}}_{i,j}}\right] \mid \mathcal{F}_\tau\right] \\
&= \mathbb{E}\left[\frac{\Psi_{t^{\mathrm{Inv}}_{i,j}}}{\Psi_\tau}\mathrm{Cov}\left(\frac{\Psi_{t^{\mathrm{Future}}_{i,j}}}{\Psi_{t^{\mathrm{Inv}}_{i,j}}}, \delta_{i,j} \mid \mathcal{F}_{t^{\mathrm{Inv}}_{i,j}}\right) \mid \mathcal{F}_\tau\right]
\end{aligned} \tag{8}$$

The term can generally become non-zero, representing the covariance between the deal's pricing error and the future level of the SDF, effectively a risk premium for the timing mismatch. Note that the final line of Equation 8 is leveraging

$$\mathbb{E}\left[\frac{\Psi_{t^{\mathrm{Future}}_{i,j}}}{\Psi_{t^{\mathrm{Inv}}_{i,j}}}\delta_{i,j} \mid \mathcal{F}_{t^{\mathrm{Inv}}_{i,j}}\right] = \mathrm{Cov}\left(\frac{\Psi_{t^{\mathrm{Future}}_{i,j}}}{\Psi_{t^{\mathrm{Inv}}_{i,j}}}, \delta_{i,j} \mid \mathcal{F}_{t^{\mathrm{Inv}}_{i,j}}\right) + \mathbb{E}\left[\frac{\Psi_{t^{\mathrm{Future}}_{i,j}}}{\Psi_{t^{\mathrm{Inv}}_{i,j}}} \mid \mathcal{F}_{t^{\mathrm{Inv}}_{i,j}}\right] \underbrace{\mathbb{E}\left[\delta_{i,j} \mid \mathcal{F}_{t^{\mathrm{Inv}}_{i,j}}\right]}_{=0 \text{ by Eq. 2}} \tag{9}$$

This result highlights a fundamental dichotomy. For *ex-ante* asset pricing tests, the Net Future Value approach is misspecified: the non-zero conditional covariance implies that the conditional moment condition $\mathbb{E}[m^{\mathrm{NFV}}_{i,j}(\theta_0) \mid \mathcal{F}_\tau] = 0$ is violated, leading to biased parameter estimates if not explicitly corrected for this risk premium. Conversely, for *ex-post* performance benchmarking, this compounding approach remains valuable. By compounding realized cash flows to a future date T, we facilitate an intuitive comparison of the fund's generated terminal wealth against a public benchmark, provided one acknowledges that the resulting *alpha* implicitly includes the compensation for the timing risk identified above. This entails that any *alpha* measured via Net Future Values is mechanically a composite metric. Specifically, we can mathematically decompose the expected NFV pricing error into two

distinct components:

$$\underbrace{\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_\tau}\delta_{i,j} \mid \mathcal{F}_\tau\right]}_{\text{Measured Alpha}} = \underbrace{\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\frac{a_{i,j}^{\text{True}}}{R_{f,t_{i,j}^{\text{Inv}}\to t_{i,j}^{\text{Future}}}} \mid \mathcal{F}_\tau\right]}_{\text{True Skill}} + \underbrace{\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\text{Cov}\left(\frac{\Psi_{t_{i,j}^{\text{Future}}}}{\Psi_{t_{i,j}^{\text{Inv}}}}, \delta_{i,j} \mid \mathcal{F}_{t_{i,j}^{\text{Inv}}}\right) \mid \mathcal{F}_\tau\right]}_{\text{Timing Risk Premium}} \tag{10}$$

where R_f is the gross risk-free return over the holding period and a^{True} is the deal's expected pricing error at inception (deal-level abnormal return, distinct from the SDF intercept α). Even if a manager has no skill ($a^{\text{True}} = 0$), the measured alpha will include the *Timing Risk Premium* which is the (discounted) covariance between the future SDF realization and the deal valuation. This is not a flaw but a feature of ex-post benchmarking: it correctly reflects that the investor was exposed to this specific path of SDF realizations. However, for ex-ante pricing tests, the presence of this non-zero *Timing Risk Premium* is problematic. Standard econometric estimators are designed to find parameters that set the average pricing error to zero. If the theoretical moment contains this positive premium, the estimator will be forced to compensate by biasing the estimated alpha downwards or distorting the SDF parameters to "explain away" the *Timing Risk Premium*.

Fund-level pricing error For the pooled, fund-level cash flow stream, we assume that the true fund valuation $V_{i,\tau}$ is generally not observable for us

$$V_{i,\tau} := \mathbb{E}\left[\sum_{t=\tau}^{T}\frac{\Psi_t}{\Psi_\tau}CF_{i,t}\,|\mathcal{F}_\tau\right] \quad \forall\ \tau \geq \min_j t_{i,j}^{\text{Inv}} \tag{11}$$

For active funds with $\tau > \min_j t_{i,j}^{\text{Inv}}$, we merely can observe reported fund NAVs as proxies for the true fund value, which are known to suffer from stale pricing and appraisal-smoothing problems (Brown et al., 2019; Korteweg, 2019; Jenkinson et al., 2020)[3]. Additionally, we define the vintage year for each fund in a mathematically convenient way as its "starting year" $v_i := \text{YEAR}\left(\min_j t_{i,j}^{\text{Inv}}\right)$.

For any discounting date $\tau \leq \min_j t_{i,j}^{\text{Inv}}$, the expected fund pricing error is equal to zero.

Proposition 1. *(Price of a pooled cash flow stream at fund inception)*

$$\mathbb{E}\left[\sum_{t=\tau}^{T}\frac{\Psi_t}{\Psi_\tau}CF_{i,t}\,|\mathcal{F}_\tau\right] = 0 \tag{12}$$

if $\tau \leq \min_j t_{i,j}^{\text{Inv}}$.

[3]In the empirical section, we will treat the most recent NAV as the final distribution cash flow for non-liquidated funds to increase the sample data.

Proof. Let us first analyze the case $\tau \leq \min_j t_{i,j}^{\text{Inv}}$ by expressing the fund cash flows in term of discounted deal pricing errors using Equation 3.

$$\mathbb{E}\left[\sum_{t=\tau}^{T} \frac{\Psi_t}{\Psi_\tau} CF_{i,t} \,|\mathcal{F}_\tau\right] = \mathbb{E}\left[\sum_{j=1}^{J} \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j} \,|\mathcal{F}_\tau\right] \tag{13}$$

Then we can simply plug in Equation 5 to receive the postulated zero expected fund pricing error.

$$\mathbb{E}\left[\sum_{j=1}^{J} \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \delta_{i,j} \,|\mathcal{F}_\tau\right] = \mathbb{E}\left[\sum_{j=1}^{J} \frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau} \cdot 0 \,|\mathcal{F}_\tau\right] = 0 \tag{14}$$

□

Next, let us analyze the case $\tau > \min_j t_{i,j}^{\text{Inv}}$ for Equation 11. For any deal j with $t_{i,j}^{\text{Inv}} < \tau$, the conditional expectation $\mathbb{E}[\delta_{i,j}|\mathcal{F}_\tau]$ represents the realized abnormal return of the deal up to time τ. For a generic risky asset, this realized return is a random variable with non-zero variance. Therefore, the condition that Equation 11 equals 0 holds *ex-post* only by chance (i.e., if realized shocks cancel out exactly). Ex-ante, without knowledge of realizations, $\tau \leq \min_j t_{i,j}^{\text{Inv}}$ is the necessary condition to guarantee a zero expected pricing error.

2.2 Least-Mean-Distance estimator

In this subsection, we introduce a new SDF estimator designed to analyze the effect of different discounting dates τ on the bias and variance of the estimated SDF parameters. In the previous subsection, we demonstrated that only the fund inception date yields unconditionally unbiased parameter estimates in a NPV-style optimization formulation. However, to be able to empirically test the impact of mixing in biased NFV terms, we choose to generalize our estimation framework so we can handle both cases flexibly. With this respect, our general idea is to rather average over multiple "suitable" discounting date candidates τ than to select only one candidate for the "unconditionally optimal" discounting date.

Henceforth, we assume that the underlying transactions within a private equity fund cannot be distinguished individually, and that only the funds total (pooled) cash flows are observable. The stochastic discount factor $\Psi_{\tau,t}$ is used to calculate the time-τ "price" $P_{\tau,t,i}$ of a **single** time-t cash flow of any given PE fund i

$$P_{\tau,t,i} := \Psi_{\tau,t} \cdot CF_{i,t} = \frac{\Psi_t}{\Psi_\tau} \cdot CF_{i,t} \qquad \forall \quad \tau, t, i \tag{15}$$

with multi-period SDF $\Psi_t = \prod_{k=1}^{t} M_k$ and $\Psi_0 = 1$. As SDFs are commonly parameterized by a vector $\theta \in \mathbb{R}^p$, i.e., $\Psi_{\tau,t} \equiv \Psi_{\tau,t}(\theta)$, our goal is to find an estimation method for the optimal θ. We denote this optimal/best/true parameter vector as θ_0. We call the numerator Ψ_t the discount part of the multi-period SDF $\Psi_{\tau,t}$ (used for present value calculations) and the denominator Ψ_τ the compound part (used for future value calculations). For each fund i and all points τ within a common

fund lifetime, the empirical "pricing error" $\epsilon_{\tau,i}$ of **all** fund cash flows is calculated as time-τ "present value"

$$\epsilon_{\tau,i} := \sum_{t=0}^{T} P_{\tau,t,i} \qquad \forall \quad \tau, i \tag{16}$$

We use the terms "price", "pricing error" and "net present value" in quotation marks to acknowledge the theoretical asset pricing problem which can arise for pooled cash flows and has been described in the previous subsection.

To better analyze the impact of different discount date τ on the estimator's bias and variance, we define the (w_i-weighted) average pricing error $\bar{\epsilon}_i$ that averages over the set $\mathcal{T}_i$

$$\bar{\epsilon}_i = w_i \cdot \frac{1}{\text{card}(\mathcal{T}_i)} \sum_{\tau \in \mathcal{T}_i} \epsilon_{\tau,i} \qquad \forall \quad i \tag{17}$$

where $\mathcal{T}_i$ gives the set of discounting dates τ for fund i which is more thoroughly described below. Finally, the scalar weighting factor w_i can be (i) one divided by the fund's invested capital for equal weighting of funds, (ii) one divided by the vintage year sum of invested capital for vintage year weighting, (iii) the scalar one for fund-size weighting (since cash flows are already scaled by fund size in this case), or (iv) some macroeconomic deflator.

The role of $\mathcal{T}_i$ The set $\mathcal{T}_i$ controls which discounting dates enter the averaged pricing error $\bar{\epsilon}_i$ in Equation 17, as visualized in Figure 1. The smallest choice $\mathcal{T}_i = \{\min_j t_{i,j}^{\text{Inv}}\}$ corresponds to a classical NPV calculation and is used by both Driessen et al. (2012) and Korteweg and Nagel (2016). The largest candidate set includes all periods from inception to the present.

Averaging over multiple dates in $\mathcal{T}_i$ was originally motivated by the "exploding alpha" problem first described by Driessen et al. (2012): when the initial cash flow is very small, an arbitrarily large alpha can discount all subsequent cash flows close to zero, rendering the beta factors irrelevant and distorting estimation. Averaging over $\mathcal{T}_i$ mitigates this degeneracy.

However, as shown in Section 2.1, including future value dates in $\mathcal{T}_i$ introduces a timing risk premium bias. The bias–variance tradeoff of different $|\mathcal{T}_i|$ choices is studied empirically in the simulation study (Section 3.3).

Remark 1 (Target parameter under horizon averaging)**.** *Our LMD estimator with $\mathcal{T}_H = \{1, 2, \ldots, 12H\}$ targets the horizon-specific pseudo-true parameter $\theta_H^* = \arg\min_\theta Q_H(\theta)$, which coincides with the structural parameter θ_0 only when $H = 0$ (NPV-only discounting). For $H > 0$, the Timing Risk Premium introduces a population-level wedge $\theta_H^* \neq \theta_0$ (see Appendix A.1 for the formal definition and Appendix A for the full decomposition). We nevertheless treat horizon averaging as a finite-sample regularization device: our simulations show that the dominant source of estimation error at short horizons is severe upward bias in factor loadings arising from the interaction of idiosyncratic cash flow noise with the nonlinear estimating equations, rather than estimator variance per se. Averaging pricing errors over an extended set of compounding dates effectively controls and stabilizes this finite-sample estimation*

error. The resulting mean squared error is minimized at approximately $H \approx 12$–15 years under our baseline DGP, where the small TRP-induced population bias is more than offset by the large reduction in finite-sample estimation error.

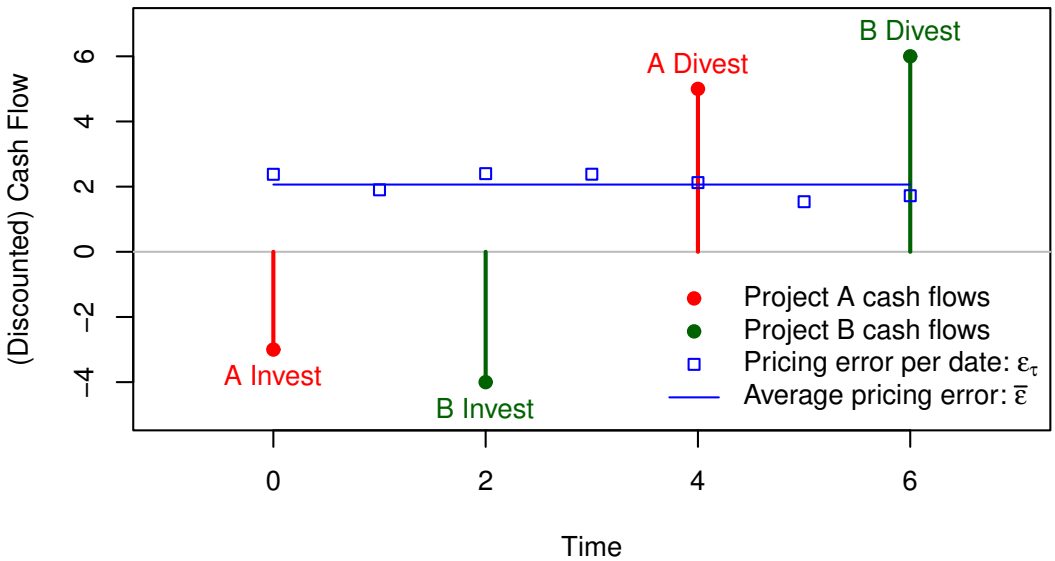

Figure 1: How to calculate and interpret the average pricing error introduced by Equation 17: The time index t is relevant for the net cash flows (black dots). The time index τ is used for the pricing error, i.e., the sum of discounted net cash flows (blue boxes). The weighted average of these "pricing errors" gives the average pricing error $\bar{\epsilon}$ as defined in Equation 17 (solid blue line). In this example, $\text{card}(\mathcal{T}_i) = 7$, i.e., the number of blue boxes.

Assumption 2. *To guarantee an unconditionally unbiased estimator, we assume to always only discount to the fund start date* $\mathcal{T}_i = \left\{\min_j t_{i,j}^{\text{Inv}}\right\}$.

Remark 2. *Assumption 2 guarantees that* $\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = 0$ *under the true parameter. For the quadratic loss* $L(x) = x^2$*, the population objective* $Q_0(\theta) = -\mathbb{E}[\bar{\epsilon}_i(\theta)^2] = -\text{Var}[\bar{\epsilon}_i(\theta)] - (\mathbb{E}[\bar{\epsilon}_i(\theta)])^2$ *involves both variance and bias terms. Consistency then requires that* θ_0 *uniquely minimizes this composite criterion (see Proposition 2).*

To find the optimal value for θ, we select an estimator from the broad class of extremum estimators.

Definition 1. *Extremum estimator (Newey and McFadden, 1994, Equation 1.1): An estimator* $\hat{\theta}$ *is an extremum estimator if there is an objective function* $Q_n(\theta)$ *such that*

$$\hat{\theta} = \arg\max_{\theta} \; Q_n(\theta)$$

for $\theta \in \Theta$ *where* Θ *is the set of all possible parameter values.*

Concretely, our Least Mean Distance (LMD) estimator (Pötscher and Prucha, 1997, Equation 7.1) minimizes the average loss of $\bar{\epsilon}$. For consistency with Definition 1, we will technically maximize the average negative loss of $\bar{\epsilon}$, which is equivalent.

$$\hat{\theta} = \arg\min_{\theta\in\Theta} \ \frac{1}{n}\sum_{i=1}^{n} L\left(\bar{\epsilon}_i\right) \equiv \arg\max_{\theta\in\Theta} \ Q_n(\theta) \quad \text{with} \quad Q_n(\theta) = -\frac{1}{n}\sum_{i=1}^{n} L\left(\bar{\epsilon}_i\right) \tag{18}$$

where $L : \mathbb{R}^1 \to \mathbb{R}^1$ denotes a loss function, e.g., $L(x) = (x-0)^2$ in the case of nonlinear least squares. Throughout the paper, the weighted average fund pricing error $\bar{\epsilon}_i \equiv \bar{\epsilon}_i(\theta)$ is regarded as nonlinear random function of the SDF parameter θ.

2.3 Asymptotic framework

By the value-additivity principle (Hansen and Richard, 1987), our SDF model holds for any aggregation level of cash flows. In practice, we form vintage-year portfolios as cross-sectional units, which mitigates idiosyncratic noise and GP financial engineering artifacts while preserving sufficient test assets for parameter identification.

Assumption 3. *For each vintage year, we pool fund cash flows to form n_v portfolios. Thus, $n = \sum_{v=1}^{V} n_v$. The boundary cases are (i) single-fund portfolios and (ii) one portfolio per vintage year. Both equal-weighted and value-weighted portfolios are considered.*

Without loss of generality, we refer to cross-sectional units as funds. The detailed comparison of aggregation levels is provided in Section 3.3.

The pricing error $\bar{\epsilon}_i$ is naturally indexed by both fund identity and vintage year, yielding a random field $\bar{\epsilon}_i \equiv \bar{\epsilon}_{i,v_i}$ that fits neither standard time-series, cross-sectional, nor panel data frameworks (cf. Figure 2). We address this by adopting a spatial framework similar to Korteweg and Nagel (2016), where they measure the economic distance between funds as a function of the cash flow overlap. In our case, it is convenient to interpret the fund vintage year v_i as the second dimension in this random field.

Yet, in this section, we mainly follow the time-series asymptotic framework of Pötscher and Prucha (1997) since our "spatial" distance measure (between vintage years) is time, and adaptation to our case is thus straightforward. If we observe only one fund per vintage year (or, equivalently, form vintage year portfolios), we will easily see that the framework of Pötscher and Prucha (1997) with time-series indexing can be applied without any major modification.

2.3.1 Vintage year asymptotics

We assume that the "spatial" (i.e., economic) distance between cross-sectional units, i.e., private equity funds/portfolios, can be measured quantitatively[4]. Our "cross-sectional" asymptotic theory lets the number of funds go to infinity $n \to \infty$. To expose our SDF to many distinct covariate realizations (economic conditions), we also want the number of vintages to increase asymptotically.

[4]Generally, the economic distance measure could include multiple dimensions, e.g., temporal, geographic, and industry sector proximity. This could be an interesting topic for future research.

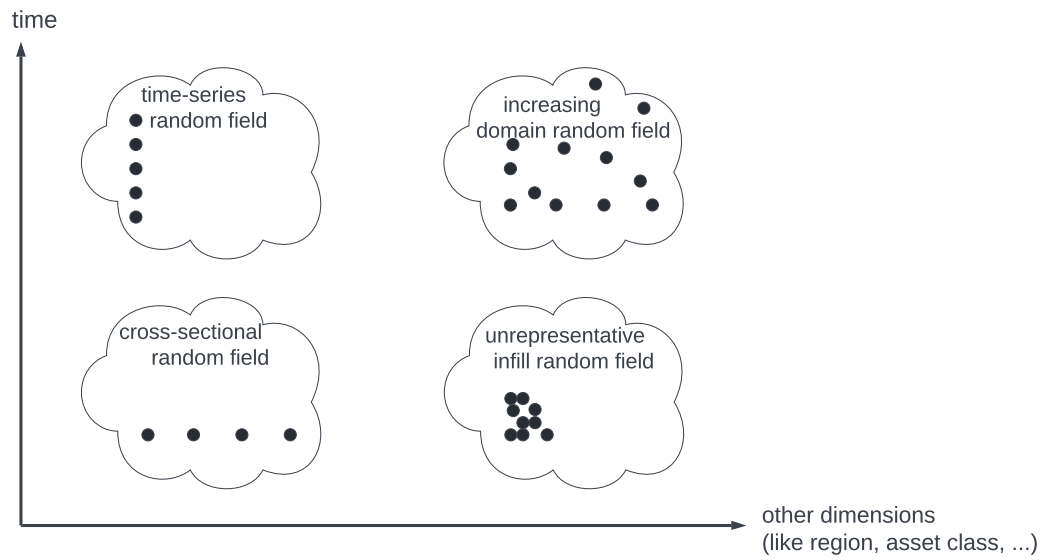

Figure 2: Visualization of generic random field types. Each black dot marks a different observation i of the cash flow data. Importantly, the time axis does **not** correspond to the index t in $CF_{i,t}$ (rather to vintage years v_i). Comparing the four choices, we want to avoid an infill random field but prefer our data to constitute an increasing domain random field. The infill random field is even asymptotically "too clustered" or better "too unrepresentative" to allow for meaningful estimation and inference. The time-series and cross-sectional random fields correspond to the standard cases in the literature but could turn out too restrictive for a general approach. By smart design (like portfolio formation), we often can map an increasing domain random field to simpler time-series or cross-sectional versions.

Assumption 4. *Vintage year asymptotics:*

1. *The number of vintage years $V \to \infty$ as $n \to \infty$.*
2. *The number of funds per vintage year is bounded by some positive constant.*[5]
3. *The maximum fund lifetime is also bounded by a positive constant.*
4. *The economic distance between fund i and j is measured by the vintage year difference $d_{i,j} = |v_i - v_j| + \rho_0 \mathbf{1}_{i \neq j}$ with minimum distance $\rho_0 > 0$. Note that two distinct funds from the same vintage year have distance $d_{i,j} = \rho_0$, reflecting our assumption that within-vintage-year dependence is captured by the SHAC kernel at the smallest lag.*

[5]This assumption is naturally satisfied when forming vintage-year portfolios (case (ii) of Assumption 3, with $n_v = 1$). For the individual fund case (i), it ensures that the SHAC estimator remains consistent and that standard errors scale with the number of vintage years rather than the total fund count. Relaxing this assumption would require cluster-robust inference at the vintage-year level.

In terms of the spatial estimation literature, this assumption postulates increasing domain asymptotics and rules out so-called infill asymptotics (cf. Figure 2). The minimum distance term ρ_0 is a means to ensure these increasing domain asymptotics (Jenish and Prucha, 2012, Assumption 1). Infill asymptotics correspond to the approach of Driessen et al. (2012), who fix the number of portfolios N and let the number of underlying funds per portfolio $n_i \to \infty$. While this yields consistency for a given set of portfolios, it fundamentally conditions on the observed macro environment: with fixed N, no new vintage yearsand hence no new factor realizationsenter the estimation. In contrast, our increasing domain framework ($V \to \infty$) permits inference that accounts for cross-vintage variation in economic conditions, which is necessary for the unconditional identification of SDF parameters (see Section 2.1). Finally, a further advantage of the *nonlinear least squares* formulation in Equation 18 is that it avoids the well-known complications arising when the number of GMM moment conditions grows with sample size (Han and Phillips, 2006; Newey and Windmeijer, 2009).

2.3.2 Law of large numbers

Under Assumption 2 (NPV-only discounting), Proposition 1 guarantees $\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = 0$. For the general case where $\mathcal{T}_i$ includes future value dates, a residual bias may persist (see Section 2.1). In both cases, the LLN ensures that the sample objective $Q_n(\theta)$ converges uniformly to its population counterpart. To approach this expected value in $\mathbb{E}[\bar{\epsilon}_i(\theta_0)]$, we rely on a spatial law of large numbers instead of applying a time-series law of large numbers. Here, we want to explicitly acknowledge the statistical dependence of pricing errors from adjacent vintage years.

Assumption 5. *Primitive conditions for uniform convergence (cf. Jenish and Prucha, 2009, Theorem 6):*

1. Near-epoch dependence (NED): *For each $\theta \in \Theta$, the random field $\{L(\bar{\epsilon}_i(\theta))\}_{i=1,\dots,n}$ is NED of size $-a$ (with $a > 0$) on an α-mixing random field $\{\eta_v\}_{v\geq 1}$ indexed by vintage years, i.e., $\|L(\bar{\epsilon}_i(\theta)) - \mathbb{E}[L(\bar{\epsilon}_i(\theta)) \mid \sigma(\eta_v : |v - v_i| \leq m)]\|_p \leq c_i \cdot \nu(m)$ with $\sup_i c_i < \infty$ and $\nu(m) = O(m^{-a})$.*
2. Mixing decay: *The base random field $\{\eta_v\}$ is α-mixing with mixing coefficients $\alpha(d) = O(d^{-\gamma})$ for some $\gamma > 2(p-1)/(p-2)$ where $p > 2$.*
3. Moment conditions: $\sup_{\theta\in\Theta} \mathbb{E}[|L(\bar{\epsilon}_i(\theta))|^p] < \infty$ *for some $p > 2$, uniformly in i.*
4. Stochastic equicontinuity: *The family $\{L(\bar{\epsilon}_i(\cdot)) : i = 1, \dots, n\}$ is stochastically equicontinuous in θ, which is satisfied if L is Lipschitz and $\bar{\epsilon}_i(\theta)$ is Lipschitz in θ uniformly in i.*

Under Assumptions 3–5, the ULLN $\sup_{\theta\in\Theta} |Q_n(\theta) - \mathbb{E}[Q_n(\theta)]| \xrightarrow{p.} 0$ follows from (Jenish and Prucha, 2009, Theorem 6). Economically, the NED condition means that the pricing error of a fund in vintage v can be well-approximated using only information from nearby vintages $\{v - m, \dots, v + m\}$, with approximation quality improving in m, which is natural since cash flow overlap decays with vintage year

distance (cf. Assumption 4). In practice, we treat funds with distance $d_{i,j} > D$ as approximately independent for the purpose of covariance estimation (cf. the bandwidth parameter $b_n = D$ in Section 2.4). Since nominal fund sizes tend to grow over time, unweighted pricing errors $\bar{\epsilon}_i$ may be non-stationary; the weighting factor w_i from Equation 17 (e.g., scaled by invested capital) normalizes the pricing errors so that the moment condition (3) is satisfied.

2.3.3 Consistency

The estimator $\hat{\theta}$ shall converge in probability to the true parameter value θ_0 as the number of distinct vintage years goes to infinity.

Assumption 6. *(Identification) The population objective $Q_0(\theta)$ has a unique maximizer at θ_0, i.e., $Q_0(\theta) < Q_0(\theta_0)$ for all $\theta \neq \theta_0$.*

Remark 3. *For $L(x) = x^2$, this requires that no alternative parameter vector $\theta \neq \theta_0$ can simultaneously achieve zero bias ($\mathbb{E}[\bar{\epsilon}(\theta)] = 0$) and equal variance ($\mathrm{Var}[\bar{\epsilon}(\theta)] = \mathrm{Var}[\bar{\epsilon}(\theta_0)]$). This is economically natural: mis-specifying the SDF factor loadings introduces systematic pricing errors whose variance exceeds the irreducible noise at θ_0. The condition is analogous to the standard rank condition in linear GMM.*

Proposition 2 (Consistency)**.** *Under Assumptions 3–6, $\hat{\theta} \xrightarrow{p} \theta_0$ as $n \to \infty$.*

Proof. We verify the four conditions of (Newey and McFadden, 1994, Theorem 2.1) for the extremum estimator $\hat{\theta} = \arg\max_{\theta \in \Theta} Q_n(\theta)$ with population objective $Q_0(\theta) := \mathbb{E}[Q_n(\theta)] = -\mathbb{E}\big[L\big(\bar{\epsilon}_i(\theta)\big)\big]$.

1. *Identification.* Since $L \geq 0$, we have $Q_0(\theta) \leq 0$ for all θ. For the case $L(x) = x^2$,
$$Q_0(\theta) = -\mathrm{Var}\big[\bar{\epsilon}(\theta)\big] - \big(\mathbb{E}[\bar{\epsilon}(\theta)]\big)^2.$$
Under Assumption 2, Proposition 1 yields $\mathbb{E}[\bar{\epsilon}(\theta_0)] = 0$, so the bias term vanishes at θ_0. For any $\theta \neq \theta_0$, at least one of (a) the bias term $(\mathbb{E}[\bar{\epsilon}(\theta)])^2 > 0$ or (b) the variance term $\mathrm{Var}[\bar{\epsilon}(\theta)] > \mathrm{Var}[\bar{\epsilon}(\theta_0)]$ must hold, ensuring $Q_0(\theta) < Q_0(\theta_0)$. This is the standard identification condition requiring that the true parameter is the unique minimizer of the expected loss.
2. *Compactness.* The parameter space Θ is compact by economic reasoning: factor loadings and intercepts are bounded by plausible ranges implied by the risk–return tradeoff.
3. *Continuity.* Continuity of $Q_0(\theta)$ follows from the continuous dependence of $\Psi_{\tau,t}(\theta)$ on θ and the dominated convergence theorem, which is a standard regularity condition.
4. *Uniform convergence.* By Assumption 5, $\sup_{\theta \in \Theta} |Q_n(\theta) - Q_0(\theta)| \xrightarrow{p} 0$, which directly satisfies the ULLN requirement of (Newey and McFadden, 1994, Section 2.1).

□

2.3.4 Central limit theorem

To assess the large-sample significance of our parameter estimates in Subsection 2.4, we establish the asymptotic distribution of $\hat{\theta}$.

Proposition 3 (Asymptotic Normality)**.** *Under the conditions of Proposition 2, suppose additionally that (i) θ_0 is interior to Θ, (ii) $Q_0(\theta)$ is twice continuously differentiable at θ_0 with nonsingular Hessian $H := \nabla_{\theta\theta} Q_0(\theta_0)$, and (iii) a spatial CLT holds for the score. Then*

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Sigma) \quad \text{as } n \to \infty,$$

with sandwich covariance matrix $\Sigma = H^{-1} \Lambda H^{-1}$, where $\Lambda := \lim_{n\to\infty} n \cdot \mathrm{Var}\big[\nabla_\theta Q_n(\theta_0)\big]$ is the long-run covariance matrix of the score.

Proof. The result follows from the general asymptotic normality theorem for extremum estimators (Newey and McFadden, 1994, Theorem 3.1). The key nonstandard ingredient is condition (iii): a CLT for the spatially dependent score $\nabla_\theta Q_n(\theta_0)$. Under the spatial near-epoch dependence structure posited in Assumption 5 and the vintage year asymptotics of Assumption 4, this follows from (Jenish and Prucha, 2012, Theorem 4), which extends the Pötscher and Prucha (1997) framework to spatial random fields. The remaining conditions of (Newey and McFadden, 1994, Theorem 3.1), i.e., interior θ_0, smooth Q_0, and stochastic equicontinuity of the score, are standard regularity conditions satisfied by the continuous dependence of $\Psi_{\tau,t}(\theta)$ on θ. □

Remark 4. *The spatial CLT of (Jenish and Prucha, 2012, Theorem 4) requires the same NED and mixing conditions as Assumption 5, supplemented by a Lindeberg-type condition on the score contributions $\nabla_\theta L(\bar{\epsilon}_i)$. The Lindeberg condition is implied by the uniform moment bound in Assumption 5(3) applied to the score.*

2.4 Large sample inference

From Proposition 3, the asymptotic covariance matrix takes the sandwich form $\Sigma = H^{-1} \Lambda H^{-1}$[6] with H and Λ as defined therein. We now describe how to construct consistent estimators $\hat{H}$ and $\hat{\Lambda}$.

We define the corresponding finite sample estimators analogously to Pötscher and Prucha (1997, Chapters 12, 13.1), and numerically approximate the first and second partial derivatives by finite differences[7]. Specifically, we use the following central difference approximations (with "small" δ) (Eu, 2017, Algorithm 2):

$$f_x(x, y) \approx \frac{f(x + \delta, y) - f(x - \delta, y)}{2\delta}$$

$$f_{xx}(x, y) \approx \frac{f(x + \delta, y) + f(x - \delta, y) - 2f(x, y)}{\delta^2}$$

[6] Since H is the Hessian of a scalar objective, it is symmetric, so $H^{-1} = (H^{-1})^\top$.

[7] As an alternative to finite differences the widespread Broyden-Fletcher-Goldfarb-Shanno (BFGS) algorithm can be applied to approximate the Hessian (Nocedal and Wright, 2006, Section 6.1).

$$f_{xy}(x,y) \approx \frac{f(x+\delta,y+\delta)+f(x-\delta,y-\delta)-f(x+\delta,y-\delta)-f(x-\delta,y+\delta)}{4\delta^2}$$

The Hessian term $\hat{H}$ is relatively straightforward

$$\hat{H} = -\frac{1}{n}\sum_{i=1}^{n} \nabla_{\theta\theta} L\left(\bar{\epsilon}_i\right)$$

The consistency $\hat{H} \xrightarrow{p} H$ is implied by the ULLN (Assumption 5) applied to the second derivatives. Due to the spatial near-epoch dependence, the involved and computationally expensive part is to consistently estimate $\hat{\Lambda}$ by a Spatial Heteroskedasticity and Autocorrelation Consistent (SHAC) covariance matrix estimator (Kim and Sun, 2011, Equation 2)

$$\hat{\Lambda} = \frac{1}{n}\sum_{i=1}^{n}\sum_{j=1}^{n} k_{i,j}\left[\nabla_\theta L\left(\bar{\epsilon}_i\right)\left(\nabla_\theta L\left(\bar{\epsilon}_j\right)\right)^\top\right] \tag{19}$$

Since $\nabla_\theta Q_n(\theta) = -\frac{1}{n}\sum_i \nabla_\theta L(\bar{\epsilon}_i)$, the score contributions are $s_i(\theta) := \nabla_\theta L(\bar{\epsilon}_i)$, and $\hat{\Lambda}$ equivalently estimates $n \cdot \mathrm{Var}[\nabla_\theta Q_n]$.

We define the kernel weight k as

$$k_{i,j} \equiv K\left(\frac{d_{i,j}}{b_n}\right)$$

with kernel function $K : \mathbb{R} \to [0,1]$ that satisfies $K(0) = 1$, $K(x) = K(-x)$, $\int_{-\infty}^{\infty} K^2(x)dx < \infty$, and $K(\cdot)$ continuous at zero and at all but a finite number of other points. A common choice is the Bartlett kernel $K_{BT}(x) = \max(0, 1 - |x|)$; see equation 2.7 in Andrews (1991) for other popular kernel choices. This means absolute vintage year differences larger than the bandwidth (or truncation) parameter $b_n = D$ are considered independent and are thus excluded from the $\hat{\Lambda}$ estimation formula.

In large samples, the vector of parameter standard errors can thus be estimated by

$$\mathrm{SE}(\hat{\theta}) = \sqrt{\mathrm{diag}\left[n^{-\frac{1}{2}}\hat{H}^{-1}\hat{\Lambda}\,\hat{H}^{-1}(n^{-\frac{1}{2}})^\top\right]} = \sqrt{\mathrm{diag}\left[\hat{H}^{-1}\hat{\Lambda}\,\hat{H}^{-1}\right]\cdot\frac{1}{n}} \tag{20}$$

The Wald test statistic for linear hypotheses $\mathcal{H}_0 : R\theta = r$ and $\mathcal{H}_1 : R\theta \neq r$ is constructed as (where $\mathcal{H}_0$ and $\mathcal{H}_1$ are hypotheses and not Hessian terms H)

$$W = (R\hat{\theta} - r)^\top \left[R\frac{\hat{H}^{-1}\hat{\Lambda}\,\hat{H}^{-1}}{n}R^\top\right]^{-1}(R\hat{\theta} - r) \overset{H_0}{\sim} \chi^2_q$$

where $\hat{\theta}$ is the $p \times 1$ parameter vector, R is a $q \times p$ matrix, and r is a $q \times 1$ vector. Usually, we select R as the $p \times p$ identity matrix, and r as $p \times 1$ vector (e.g., of zeros). Under the null hypothesis, W is chi-squared distributed with q degrees of freedom. As large values of W indicate the rejection of H_0, the corresponding p-value is calculated as $1 - F_{\chi^2_q}(W)$ where $F_{\chi^2_q}$ is the cumulative distribution function of a chi-squared random variable with q degrees of freedom.

However, given the limited amount of available private equity data (typically the oldest vintages start in the 1980s), asymptotic characterizations of Σ and $\mathrm{SE}(\hat{\theta})$ are of limited importance. In empirical applications, the small sample behavior of an estimation method for private equity data is more relevant than its asymptotic theory. The standard asymptotic distribution associated with an estimator is generally not valid for post-model-selection inference, i.e., if a model selection procedure is applied to find the best model from a collection of competitors (Leeb and Pötscher, 2005).

2.5 Comparison to similar estimators

Our Least-Mean-Distance (LMD) estimator introduced in Section 2.2 belongs to the class of semiparametric nonlinear M-estimators as defined in Pötscher and Prucha (1997) which are extremum estimators. To gain more flexibility and avoid unneeded complexity, we intentionally opt against the most prominent semiparametric nonlinear M-estimator framework, i.e., classical time-series Generalized Method of Moments (GMM) (Hansen, 1982, 2012). A classical GMM approach requires the construction of stationary, ergodic time-series of moment conditions that are used to empirically estimate the expected value of pricing errors in Equation 16. The stationarity requirement of classical time-series GMM limits (i) more elaborate weighting schemes for w, like fund-size weighting, and (ii) the usage of fund cash flows from non-realized vintages.

The flexibility of our framework directly embeds the two most prominent asset pricing approaches in the private equity literature as special cases. We formalize this relationship below to demonstrate that our proposed methodology recovers both the cross-sectional approach of Driessen et al. (2012) and the time-series approach of Korteweg and Nagel (2016) as special cases under specific aggregation and loss-function choices.

Proposition 4 (Relationship to existing estimators)**.** *The LMD estimator defined in Equation 18 recovers the following two estimators as special cases:*

1. ***Driessen et al. (2012)****: By restricting the set of discounting dates to fund inception ($\mathcal{T}_i = \{\min_j t_{i,j}^{\mathrm{Inv}}\}$), applying equal weights ($w_i = 1$), and choosing a quadratic loss function ($L(x) = x^2$), the LMD minimization objective becomes identical to their cross-sectional Generalized Method of Moments (GMM) objective function.*
2. ***Korteweg and Nagel (2016)****: By interchanging the order of the summation and the loss function evaluation in the LMD minimization objective (i.e., evaluating $L\left(\frac{1}{n}\sum_{i=1}^{n}\bar{\epsilon}_i\right)$ instead of $\frac{1}{n}\sum_{i=1}^{n}L(\bar{\epsilon}_i)$), while defining the loss function as the standard quadratic GMM objective ($L(x) = x^\top W x$) with weighting matrix W and pooling all funds into a single cross-sectional unit ($n = 1$), our framework yields their time-series GMM estimator.*

This result establishes a structural connection at the objective-function level; it does not imply that the estimators are statistically equivalent in finite samples, as they differ in data inputs, conditioning, and identification strength.

2.5.1 Comparison to Driessen et al. (2012)

The Driessen et al. (2012) approach is most closely related to our methodology. One important difference is that we select a simpler and more flexible LMD estimator instead of a cross-sectional GMM approach. In our view, the choice of the more complex cross-sectional GMM just introduces conceptual complexity without changing the estimating equations since the underlying formulas are basically the same as for our LMD estimator[8].

Proof of Proposition 4 (Part 1). Applying the restrictions $\mathcal{T}_i = \{\min_j t_{i,j}^{\text{Inv}}\}$ and $w_i = 1$ to our average pricing error from Equation 17 yields exactly the inception-based net present value:

$$\bar{\epsilon}_i = \sum_{t=0}^{T} \frac{\Psi_t}{\Psi_{\min_j t_{i,j}^{\text{Inv}}}} \cdot CF_{i,t} = PV_i$$

Assuming equal weights for all funds leads to the minimization formulation of the LMD estimator in Equation 18 with quadratic loss $L(x) = x^2$:

$$\hat{\theta} = \arg\min_{\theta} \frac{1}{n} \sum_{i=1}^{n} (PV_i)^2 \tag{21}$$

This minimization objective is equivalent to the inner product of the pricing error vector with itself, which precisely recovers Equation 3 in Driessen et al. (2012): $\min_\theta e'e$, where e is an $n \times 1$ vector containing the pricing errors of the n cross-sectional units. □

Consequently, if someone accepts the assumptions from Subsection 2.3, our large sample inference framework from Subsection 2.4 applies to their case without any significant modification.

As a first limitation, they have to regard vintage-year portfolios as their cross-sectional units; we can also use individual funds. In this context, Driessen et al. (2012) state that "to identify β, it is essential that the different FoFs are exposed to different market returns." We note that their estimator is valid as a cross-sectional approach: identification requires only that the cross-section of funds provides sufficient variation in factor exposure at a given market return realization[9]. That said, having FoFs exposed to different market environments (i.e., different vintage years) does improve statistical power and is the basis for our increasing domain asymptotics.

Second, the Driessen et al. (2012) asymptotic theory is based on infill asymptotics: they fix the number of vintage year portfolios n and let the number of underlying funds per portfolio $n_i \to \infty$. As discussed in Section 2.3, this conditions inference on the observed macro environment since no new vintage years enter the estimation. Since each portfolio constitutes a moment condition from the GMM perspective, standard GMM theory requires n to remain fixed (Han and Phillips, 2006; Newey and Windmeijer, 2009). In contrast, our increasing domain framework

[8]The formulas are this similar because Driessen et al. (2012) use the identity matrix as GMM weighting matrix and skip the second GMM step.

[9]In a classic cross-sectional regression, we only have one market return realization.

lets the number of vintage years $V \to \infty$ while bounding the number of funds per vintage, enabling inference that reflects cross-vintage variation in economic conditions.

Further, Driessen et al. (2012) discount all fund cash flows exclusively to the fund inception date, corresponding to $\mathcal{T}_i = \{\min_j t_{i,j}^{\text{Inv}}\}$ in our notation. Under Assumption 2, this is also the only choice that yields unconditionally unbiased estimates. Our framework generalizes this by allowing $\mathcal{T}_i$ to contain additional discounting dates, which we use to empirically study the bias–variance tradeoff analyzed in Section 2.1. Finally, Driessen et al. (2012) apply simple cross-sectional bootstrapping to obtain standard errors; in contrast, in Subsection 3.2, we complement our SHAC-based asymptotic inference with hv-block cross-validation (Racine, 2000) as a stability diagnostic and model-selection criterion adapted to the near-epoch dependence of PE fund data.

2.5.2 Comparison to Korteweg and Nagel (2016)

Korteweg and Nagel (2016) were the first to employ a spatial framework for asymptotic inference on private equity fund data. They measure the economic distance between funds by the degree of cash flow overlap and use the spatial HAC estimator of Conley (1999) to construct robust standard errors. Our spatial HAC framework similarly accounts for cross-vintage dependence but draws on the near-epoch dependent theory of Pötscher and Prucha (1997); Kim and Sun (2011); Jenish and Prucha (2012). A key methodological difference is that Korteweg and Nagel (2016) estimate the SDF on synthetic public market portfolios using a time-series GMM estimator and then apply the resulting model to evaluate PE cash flows, whereas we estimate the SDF directly from PE fund cash flows. Korteweg and Nagel (2016) choose this procedure since "tests that use the ex post equity premium as a performance benchmark should be more powerful than tests that use the ex ante equity premium."

Proof of Proposition 4 (Part 2). The time-series GMM objective is formed by pooling the expected values of the moment conditions. If we treat each of the n cross-sectional units equivalently as a moment condition and compute the average pricing error across the panel, we obtain the m-dimensional sample moment $\frac{1}{n}\sum_{i=1}^{n} \bar{\epsilon}_i \in \mathbb{R}^m$. The standard GMM estimator minimizes the quadratic form of these moments weighted by a matrix W. Formally, defining $L(x) = x^\top W x$, the time-series GMM objective evaluated on the pooled sample moments becomes:

$$Q_{GMM}(\theta) = L\left(\frac{1}{n}\sum_{i=1}^{n} \bar{\epsilon}_i\right) = \left(\frac{1}{n}\sum_{i=1}^{n} \bar{\epsilon}_i\right)^\top W \left(\frac{1}{n}\sum_{i=1}^{n} \bar{\epsilon}_i\right) \tag{22}$$

Setting $W = I$ corresponds exactly to Equation 18 in Korteweg and Nagel (2016). Therefore, the minimization formulation of our LMD objective in Equation 18 (i.e., $\frac{1}{n}\sum_{i=1}^{n} L(\bar{\epsilon}_i)$) differs from Korteweg and Nagel (2016) solely by whether the loss function $L(\cdot)$ is evaluated before or after pooling the cross-sectional errors. □

Time-series GMM estimators inherently bear the risk of under-identification if the corresponding time-series is constructed by pooling all fund cash flows from a

given fund type. When PE fund cash flows are pooled into a single moment condition as in Equation 22 with $m = 1$, the resulting estimator has just one moment condition, which is insufficient to identify multi-parameter models[10]. Note that Korteweg and Nagel (2016) avoid this issue by estimating the SDF from public market data rather than PE cash flows; the under-identification concern arises only if one applies their framework directly to PE data, as we do in our estimator. To counter under-identification, additional characteristic-based fund portfolios could be formed to increase the number of moment conditions per fund type; also, random portfolios combined with bootstrapping could make sense. Yet, Korteweg and Nagel (2016) take another approach and introduce the concept of Generalized Public Market Equivalent (GPME), which elegantly avoids the under-identification issue. Firstly, a public market SDF model is estimated by pricing public trading strategies that shall replicate PE funds instead of directly pricing the observed PE fund cash flows. Only in a second step are these models applied to evaluate private equity fund cash flows.

In summary, our LMD estimator extends Driessen et al. (2012) by (a) permitting flexible discounting dates via $\mathcal{T}_i$, (b) employing increasing domain rather than infill asymptotics, and (c) providing SHAC-based inference. Relative to Korteweg and Nagel (2016), we directly price PE cash flows rather than pricing public replicating strategies, which avoids potential misspecification of the public SDF but requires stronger identification assumptions from the PE data alone. Table 2 summarizes the most prominent distinctions between the three approaches.

	Driessen et al. (2012)	Korteweg and Nagel (2016)	Our approach
Estimator type	Cross-sectional NLS (one-step GMM)	Time-series GMM (public SDF)	Nonlinear Least-Mean-Distance
Cash flows priced	PE fund cash flows	Public replicating strategies	PE fund cash flows
Discounting dates	Fund inception only	Fund inception only	Flexible via $\mathcal{T}_i$
Asymptotics	Infill ($n_i \to \infty$, N fixed)	$V \to \infty$	Increasing domain ($V \to \infty$, n_v bounded)
Inference	Bootstrap	Spatial HAC	Spatial HAC
Cross-sectional unit	Vintage year portfolio	Single fund	Both (tested empirically)
SDF specification	Linear factor model	Exponentially affine	Both (tested empirically)

Table 2: Comparison of SDF estimation frameworks for private equity.

[10]In contrast, our estimator corresponds to the opposite edge case with asymptotically an infinite number of LMD "moment conditions" (units to price) as we let $n \to \infty$.

3 Empirical application

3.1 Data

We use the Preqin cash flow data set as of 14th April 2022, which is well known in the academic private equity literature (Harris et al., 2014; Brown et al., 2015; Korteweg and Nagel, 2016; Ang et al., 2018). To keep the empirical analysis clear and concise, we filter the dataset for Buyout (BO) and Venture Capital (VC) funds. We only keep funds with fund currency US Dollar (USD) and region equal to North America.

For this subset of BO and VC funds, we extract all (i) equal-weighted and (ii) value-weighted (or fund-size-weighted) cash flow series. In the overall dataset, the minimum vintage year is 1982, and the maximum is 2021. We explicitly refrain from excluding the most recent vintage years. Section 4.1 assess the impact of unseasoned fund NAVs on our empirical estimation results. On this basis, we decide how many recent vintage years we want to remove to reduce reliance of unrealized NAVs. For the vintage years, we keep, all liquidated and non-liquidated funds enter our estimation. For non-liquidated funds, we treat the latest net asset value as final cash flow.

The public market factors that enter our SDF draw on the US dataset of the recently popularized q^5 investment factor model sourced from `http://global-q.org/factors.html` (Hou et al., 2015, 2021). Their five-factor model includes the market excess return (MKT), a size factor (ME), an investment factor (IA), a return on equity factor (ROE), and an expected growth factor (EG). The maximum date in the q^5 investment factor dataset is 2024-12-31.

3.2 Model and estimator specifications

We test a simple linear SDF model similar to Driessen et al. (2012)

$$\Psi^{\mathrm{SL}}_{\tau,t}(\theta) = \frac{\prod_{h=0}^{\tau} \ (1 + \alpha + r_{f,h} + \sum_m \ \beta_m \ F_{m,h})}{\prod_{h=0}^{t} \ (1 + \alpha + r_{f,h} + \sum_m \ \beta_m \ F_{m,h})} \tag{23}$$

and an exponential affine SDF model adapted from Korteweg and Nagel (2016)

$$\Psi^{\mathrm{EA}}_{\tau,t}(\theta) = \exp\left[-\sum_{h=\tau+1}^{t}\left(\alpha + \log(1 + r_{f,h}) + \sum_{m\in M} \beta_m \cdot \log(1 + F_{m,h})\right)\right] \tag{24}$$

with (arithmetic) risk-free return $r_f = R_{rf} - 1$, (arithmetic) zero-net-investment portfolio returns F_m, and parameter vector $\theta = (\alpha, \beta)$. To avoid overfitting, we restrict our analysis to six simple SDF models that contain {MKT} alone or {MKT} plus {ME or IA or ROE or EG or Alpha}. In Equation 18, we use the quadratic loss function $L(x) = x^2$.

To assess the parameter significance, we compute the asymptotic standard errors as outlined in Subsection 2.4. For the Bartlett kernel's bandwidth $b_n = D$ we select 12 years, i.e., funds with absolute vintage year differences larger than 12 years are assumed to be independent.

Additionally, we assess the out-of-sample stability and model-selection properties of our SDF models. To account for the dependency between funds from adjacent vintage years caused by overlapping fund cash flows, we draw on hv-block cross-validation (Racine, 2000). As Racine (2000) demonstrates, hv-block cross-validation is a consistent model-selection criterion for dependent data: the h-block gap between training and validation sets ensures approximate independence, while the v-block provides a sufficiently large validation sample within each fold. In our implementation, we form three partitions for several vintage year groups. The validation set (v-block) always contains funds of three neighboring vintage years (e.g., 2000, 2001, 2002). To reduce the dependency between training and validation set, we remove all funds from three-year-adjacent vintage years, i.e., the h-block (e.g., 1997, 1998, 1999, 2003, 2004, 2005). Funds from the remaining vintage years enter the training set and are thus used for model estimation (e.g., 1985-1996, 2006-end). The number of validation folds adapts dynamically to the selected maximum vintage year cutoff in any given empirical estimation run. For instance, when estimating the full sample through 2021, we apply ten-fold cross-validation; when estimating a heavily seasoned sample truncated at 2010 to avoid reliance on subjective interim NAVs (see Section **??**), the framework naturally truncates to seven active folds. Table 3 illustrates the full ten partitions used for the unrestricted data set.

Remark 5 (Cross-validation as stability diagnostic)**.** *The hv-block cross-validation quantities reported throughout this paper serve as* ***stability diagnostics and model-selection criteria*** *rather than inferential standard errors. They summarize the fold-to-fold variation in parameter estimates under dependence-reducing sample splits and are informative about model sensitivity and effective parameter curvature, but they do not provide a valid null distribution for the full-sample estimator. For formal statistical inference (standard errors and hypothesis tests), we rely on the SHAC-based asymptotic framework of Section 2.4. In practice, hv-block cross-validation complements the asymptotic analysis by (i) diagnosing model stability and out-of-sample performance without relying on asymptotics and (ii) flagging over-parameterized specifications that appear well-identified in-sample but collapse out-of-sample.*

Remark 6 (Definition of Horizon)**.** *Throughout the empirical section, we use the term* Horizon H *to denote the maximum compounding horizon in years used in the set of discounting dates $\mathcal{T}$. Formally, assuming a contiguous set of monthly discounting dates $\mathcal{T} = \{1, 2, \ldots, m_{\max}\}$ (where $m_{\max}$ is the maximum month), the Horizon is defined as $H = m_{\max}/12$. Consequently, estimating a model with a Horizon H implies that the estimator minimizes the average pricing error over the set of discounting dates $\mathcal{T}_H = \{1, \ldots, 12 \cdot H\}$, formally denoted as $\bar{\epsilon}_i^{(H)}$. Note that the Horizon has as upper bound the max cash flow date per fund or portfolio in our empirical application.*

Remark 7 (Alpha Bounds)**.** *In our optimization procedure, we limit the Alpha factor α to the range of $[-2\%, +2\%]$ per month to avoid the "exploding alpha" issue documented in (Driessen et al., 2012, Footnote 8): "Note that one needs to put an upper bound on α when performing the optimization in equation (3). This is to exclude the possibility that α tends to infinity." In this degenerate scenario, an*

infinite alpha can almost perfectly price any cash flow stream, rendering the beta risk factors irrelevant.

training.before	h-block.before	v-block	h-block.after	training.after
estimation	remove	validation	remove	estimation
start-1984	1985,1986,1987	1988,1989,1990	1991,1992,1993	1994-end
start-1987	1988,1989,1990	1991,1992,1993	1994,1995,1996	1997-end
start-1990	1991,1992,1993	1994,1995,1996	1997,1998,1999	2000-end
start-1993	1994,1995,1996	1997,1998,1999	2000,2001,2002	2003-end
start-1996	1997,1998,1999	2000,2001,2002	2003,2004,2005	2006-end
start-1999	2000,2001,2002	2003,2004,2005	2006,2007,2008	2009-end
start-2002	2003,2004,2005	2006,2007,2008	2009,2010,2011	2012-end
start-2005	2006,2007,2008	2009,2010,2011	2012,2013,2014	2015-end
start-2008	2009,2010,2011	2012,2013,2014	2015,2016,2017	2018-end
start-2011	2012,2013,2014	2015,2016,2017	2018,2019,2020	2021-end

Table 3: Partitions used for hv-block cross-validation. The number of active folds truncates dynamically based on the maximum vintage year cutoff of the specific estimation run.

3.3 Simulation study

Our Monte Carlo experiments examine the bias and variance of our estimation methodology in finite samples.[11] Specifically, we investigate five key dimensions of the estimation framework: (i) the finite-sample efficiency gains of forming vintage-year portfolios versus using individual funds; (ii) the tradeoff between expanding the time-series dimension (number of vintage years) and the cross-sectional dimension (funds per vintage); (iii) the robustness of the estimates to the functional specification of the SDF (simple linear vs. exponential affine); (iv) the behavior of multivariate identification under idiosyncratic noise using two-factor models; and (v) as our main overarching topic, the optimal choice of the maximum compounding Horizon (i.e., the set of discounting dates $\mathcal{T}$).

We use historical q-investment factors from 1986 to 2005 and simulate 20 funds for each of these 20 vintage years. This means we condition our simulation analysis on realized q-investment factors paths (see the Macro-Conditioning Kernel discussion in Section 2.1). Each fund contains 15 deals with equal investment amounts and exactly one divestment cash flow. Deals are entered within the first five years of the fund lifetime following a discrete uniform distribution and afterward held between one to ten years again uniformly distributed. The deal returns are generated by the simple linear or exponential affine SDF models described in Equations 23 and 24. In the base case, we just use the MKT factor with $\beta_{\text{MKT}} = 1$ and in each

[11] As with any simulation study, it investigates the ability to identify the assumed data generating process rather than validating the corresponding SDF model itself.

month, add a normal i.i.d. error term with monthly standard deviation $\sigma = 0.2$ (annualized variance $12 \cdot 0.2^2 = 0.48$, corresponding to an annualized volatility of $0.2\sqrt{12} \approx 69\%$) and zero mean. In the exponential affine case, we adjust the lognormally distributed error mean to zero by subtracting $0.5\sigma^2$. If a negative return exceeds -100%, the company defaults with a zero exit cash flow. In contrast, the error term in the simulations of Driessen et al. (2012) is more well-behaved as it follows a shifted lognormal distribution that, even with arbitrarily high error term variance, just allows for returns below say -99%, if the market return is close to its lower bound (see equation 9 in their online appendix). We specifically analyze the remaining divergence from their simulation design in Appendix B.1. In our simulation scenarios, we evaluate the estimator across a full grid of compounding Horizons, where the maximum month in the set of discounting dates $\mathcal{T}$ ranges from 0 up to 30 years (0-360 months). To assess our estimator's bias and variance, we simulate 1000 test scenarios for both vintage-year and single-fund portfolios.

Cross-sectional unit i: As anticipated in Subsection 2.3, vintage year portfolios demonstrate lower bias and variance than individual funds. Under the simple linear SDF and a 15-year Horizon (maximum month 180), the mean (standard deviation) of the coefficient estimate $\hat{\beta}_{\text{MKT}}$ is 0.966 (0.188) for vintage year portfolios, compared to 0.887 (0.359) for individual funds. The superiority of the portfolio approach is especially striking when evaluating the entire Horizon curve in Figure 3. Estimates for individual funds are highly unstable, exhibiting severe upward bias at short Horizons ($\hat{\beta}_{\text{MKT}} \approx 1.5$) and deteriorating to negative values at long Horizons. In contrast, vintage year portfolios yield substantially more stable estimates that remain within a narrow, economically plausible range around the true value of 1.0. The portfolio approach consistently achieves a lower standard deviation across all simulated Horizons, confirming that grouping funds effectively mitigates idiosyncratic noise.

This trade-off between bias/variance reduction and sample size has important implications. While forming vintage year portfolios substantially improves the accuracy of the estimates, it dramatically reduces the number of cross-sectional units, thereby weakening the relevance of asymptotic approximations. This constraint likely explains the choice of Korteweg and Nagel (2016) to rely on individual funds as cross-sectional units in their asymptotic SHAC framework, enabling them to exploit a larger sample size for smaller standard errors.

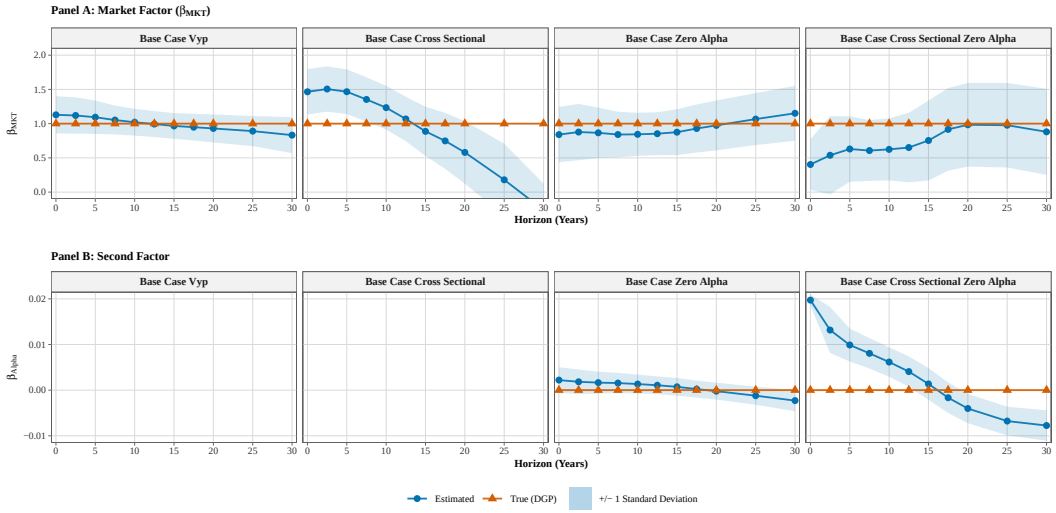

Figure 3: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Single-fund vs. vintage-year portfolios.

Varying vintages V and portfolio sizes n/V: A fundamental question in private equity panel data econometrics is whether precision is best improved by extending the time-series dimension (V, number of vintage years) or the cross-sectional dimension (n/V, number of funds per vintage portfolio). To assess this tradeoff, we compare five data configurations (depicted in Figures 4 and 5): (1) our base case for this experiment ($V = 20$, $n/V = 40$); (2) an extended vintage sample with small portfolios ($V = 39$, $n/V = 10$); (3) an extended vintage sample with medium portfolios ($V = 39$, $n/V = 20$); (4) an early-vintage sub-sample ($V = 10$, years 1986–1995, $n/V = 20$); and (5) a late-vintage sub-sample ($V = 10$, years 1996–2005, $n/V = 20$).

Our simulation yields a clear prescriptive result for empirical practice: increasing the number of cross-sectional units per portfolio (n/V) is substantially more effective at suppressing estimator variance than adding more vintage years (V). For the single-factor model at the 15-year Horizon, transitioning from 10 funds to 20 funds per vintage over a 39-year period (scenarios 2 vs. 3) reduces the standard deviation of $\hat{\beta}_{\text{MKT}}$ by roughly 22%, from 0.218 to 0.169. In contrast, shrinking the time-series dimension from 39 years to just 10 years (while holding $n/V = 20$ constant) elevates the variance only moderately for the early-vintage sub-sample ($\sigma(\hat{\beta}_{\text{MKT}}) \approx 0.194$ for 1986–1995), though the late-vintage sub-sample (1996–2005) exhibits considerably higher variance ($\sigma(\hat{\beta}_{\text{MKT}}) \approx 0.432$), reflecting the sensitivity to the realized macro environment. This hierarchy of variance reduction is even more pronounced in the two-factor setting, where idiosyncratic cash flow noise severely impairs multivariate identification: increasing n/V aggressively dampens the spurious α variance. Consequently, our findings provide a compelling rationale for the finite-sample superiority of the Driessen et al. (2012) infill asymptotics (fixing V, letting $n_v \to \infty$) over approaches that rely purely on time-series aggregation.

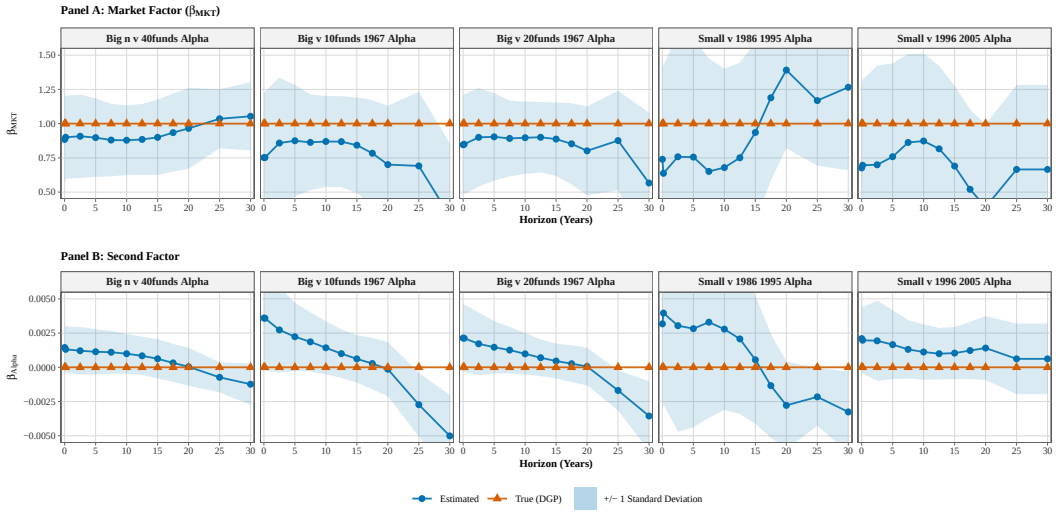

Figure 4: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Varying vintages V and portfolio sizes n/V.

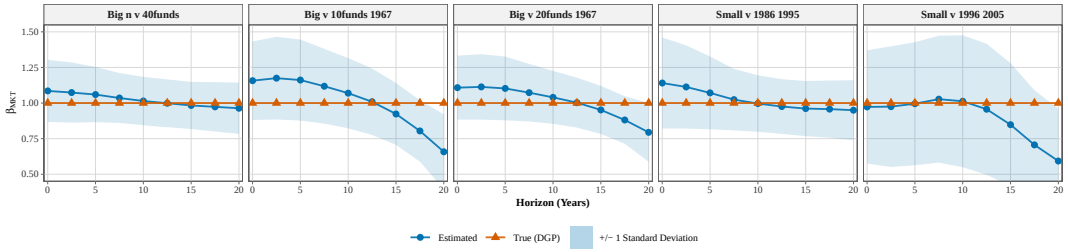

Figure 5: **Simulation ($\beta_{\mathbf{MKT}}$):** Varying vintages V and portfolio sizes n/V.

SDF model Ψ: In Figure 6, we compare the simple linear SDF model against the exponential affine specification. For the single-factor model, the bias is minimized at the 12.5-year Horizon (150 months), where both functional forms perform almost identically and yield estimates extremely close to the true value of 1.0: the exponential affine yields a mean market beta of 0.999 (0.182), matching the simple linear's 0.994 (0.189). For the two-factor (MKT+Alpha) model at this same 12.5-year Horizon, both functional forms again yield nearly identical parameter results, though with an attenuated market beta ($\hat{\beta}_{\text{MKT}} \approx 0.85$) and a minor spurious positive alpha ($\hat{\alpha} \approx 0.001$). In the two-factor case, the market beta bias is actually minimized later, around the 20-year Horizon ($\hat{\beta}_{\text{MKT}} \approx 0.97$, $\hat{\alpha} \approx -0.0002$).

However, a salient difference emerges at extremely short compounding Horizons. At a 1-month Horizon (resembling an NPV-only discounting framework), the exponential affine formulation deteriorates much more severely in both models, manifesting a larger upward bias (e.g., in the single-factor case, $\hat{\beta}_{\text{MKT}} = 1.21$ vs. 1.13) and elevated variance (0.39 vs. 0.27) compared to the linear model. Overall, the perceived superiority of exponential affine SDFs (Gourieroux and Monfort, 2007;

Bertholon et al., 2008) does not translate into empirical finite-sample advantages in our framework. The simple linear model actually appears slightly better behaved.

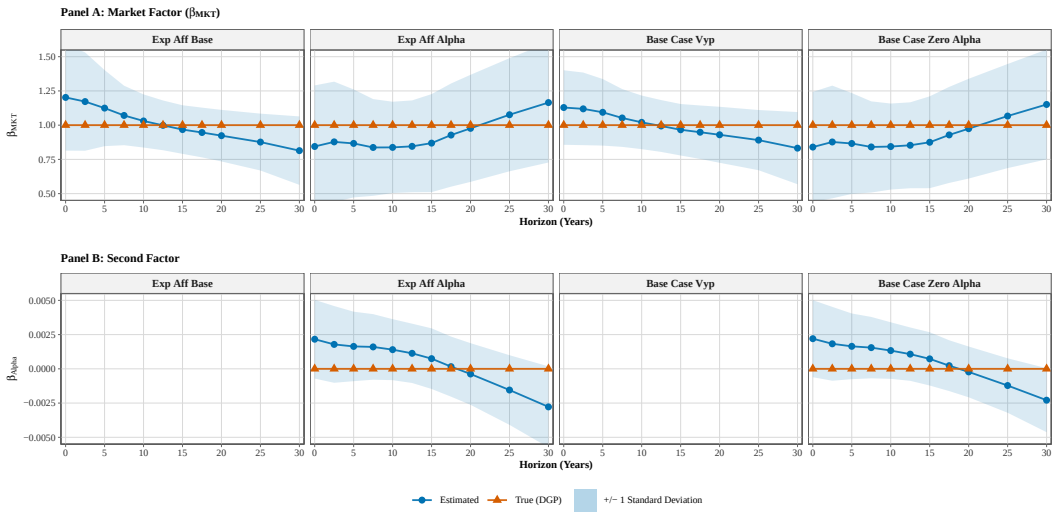

Figure 6: **Simulation ($\beta_{\mathbf{MKT}}$, α):** Exponential affine vs. linear models.

Two-factor models (q^5 factors): Figure 7 reports simulation results for two-factor models that pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG), with true loadings $\beta_{\text{MKT}} = 1$ and $\beta_{\text{2nd}} = 0.5$. A common pattern emerges across all four specifications: the second-factor loading is substantially overestimated at short Horizons and attenuated toward (or beyond) the true value at long Horizons, while the market factor absorbs a compensating bias in the opposite direction. Estimation variance is markedly larger than in the single-factor case, reflecting the collinearity between correlated factors in a limited cross-section of vintage year portfolios. Table 4 summarizes the key estimates at the shortest Horizon and at 15 years, which is approximately the MSE-minimizing Horizon for most models; we highlight the most notable factor-specific features below.

The ME model is the only specification in which $\hat{\beta}_{\text{2nd}}$ undergoes a sign change (at roughly 17.5 years), declining from 0.86 to -0.02 over the full Horizon range, a severe attenuation that would qualitatively alter economic conclusions. The IA model exhibits a large second-factor bias at short Horizons ($\hat{\beta}_{\text{IA}} \approx 1.06$ vs. truth 0.5) but both loadings converge close to their true values near the 17.5-year Horizon ($\hat{\beta}_{\text{MKT}} = 0.97$, $\hat{\beta}_{\text{IA}} = 0.52$). The EG model exhibits the largest overall estimation variance ($\sigma(\hat{\beta}_{\text{EG}}) \approx 1.5$ at the shortest Horizon) but achieves a reasonably accurate joint estimate around 15–17.5 years ($\hat{\beta}_{\text{MKT}} = 0.91$–$1.02$, $\hat{\beta}_{\text{EG}} = 0.47$–$0.56$). The ROE model is the exception: both loadings remain persistently biased at all Horizons, with $\hat{\beta}_{\text{MKT}}$ never exceeding 0.91 and $\hat{\beta}_{\text{ROE}}$ never falling below 0.62. These results suggest that empirical researchers should exercise caution when interpreting two-factor SDF estimates for private equity, and that the MSE-minimizing Horizon is model-dependent, typically falling in the 12.5–17.5-year range.

Model	$\hat{\beta}_{\text{MKT}}$		$\hat{\beta}_{\text{2nd}}$		$\sigma(\hat{\beta}_{\text{MKT}})$		$\sigma(\hat{\beta}_{\text{2nd}})$	
	0 yr	15 yr	0 yr	15 yr	0 yr	15 yr	0 yr	15 yr
ME + MKT	1.15	0.97	0.87	0.59	0.28	0.21	0.46	0.40
IA + MKT	0.97	0.92	1.06	0.64	0.34	0.30	0.74	0.53
ROE + MKT	0.88	0.83	0.98	0.69	0.43	0.49	0.61	0.49
EG + MKT	0.87	0.91	0.99	0.56	0.61	0.45	1.53	0.31

True values: $\beta_{\text{MKT}} = 1$, $\beta_{\text{2nd}} = 0.5$.

Table 4: Two-factor simulation summary: mean estimates and standard deviations at the shortest (0 yr) and the 15-year Horizon.

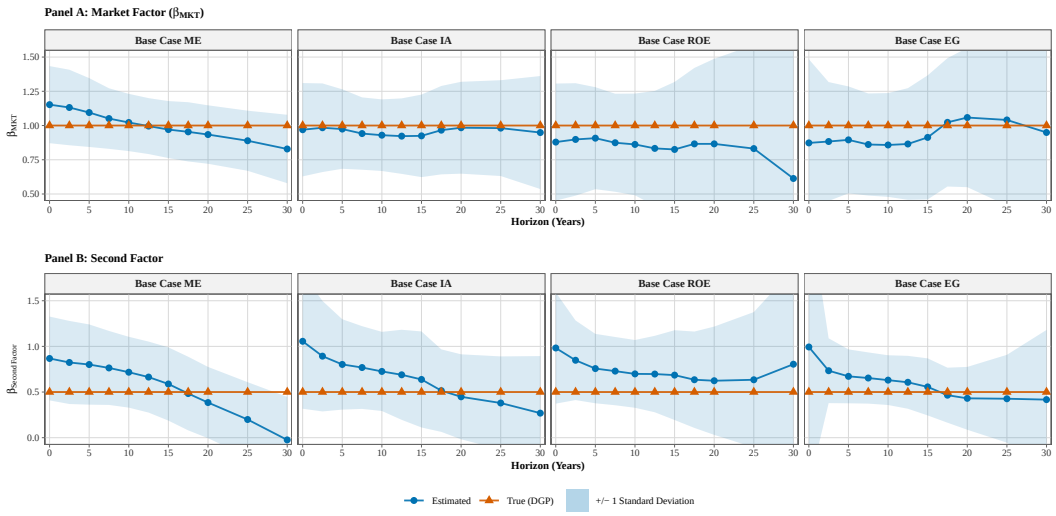

Figure 7: **Simulation (q^5 factors):** Two-factor models using VYP.

Size of set $\mathcal{T}$ (Maximum Compounding Horizon): Our most critical simulation finding relates to the optimal choice of the maximum compounding Horizon, i.e., the size of the set of discounting dates $\mathcal{T}$. From a purely theoretical perspective (cf. Section 2.1), discounting all cash flows exclusively to the fund inception date, corresponding to a zero-month or NPV-only Horizon in our notation, should yield unconditionally unbiased estimates under Assumption 2. This minimal choice for $\mathcal{T}$ corresponds exactly to the established Driessen et al. (2012) framework.

However, our diverse simulation results reveal a puzzling disconnect between asymptotic theory and finite-sample performance: extending the set of discounting dates deep into the fund lifecycle drastically improves the estimator's precision. Across almost all model specifications (see Figures 7 and 6), bias is minimized not at the theoretical 0-year Horizon, but rather around a 12.5 to 15-year compounding Horizon. At extremely short Horizons, idiosyncratic cash flow noise interacts non-linearly with the estimation procedure, inducing severe upward bias in factor loadings. By averaging pricing errors over multiple future dates, effectively incor-

porating the Timing Risk Premium, our approach acts as a powerful finite-sample bias correction.

Interestingly, this "variance reduction technique" improves the mean estimate far more than it reduces standard errors. While estimator variance traces a shallow U-shape with a minimum near the 15-year Horizon (see Table 4), this reduction is secondary to the striking stabilization of the parameter bias. We further analyze this puzzling result from a theoretical perspective in Appendix A, detailing the *bias–variance tradeoff* and the role of the Timing Risk Premium.[12]

Simulation conclusion: To conclude, our simulation study rationalizes two key practices from the Driessen et al. (2012) paper: (i) vintage-year portfolio formation helps to improve estimator precision, and (ii) increasing the number of funds per vintage seems more effective in controlling estimator variance than increasing the number of vintages (for i.i.d. error terms). Our second finding provides an empirical rationale for the choice of Driessen et al. (2012) to employ infill asymptotics that let the number of underlying funds per vintage tend to infinity. However, our correctly specified scenarios cannot support the assumption of Korteweg and Nagel (2016) that (iii) the exponential affine SDF is superior to the simple linear SDF in a multi-period framework; in fact, their finite-sample bias and variance profiles are remarkably similar. Our simulation study also suggests that (iv) averaging pricing errors over multiple dates substantially reduces the severe short-Horizon bias inherent to the original procedure of Driessen et al. (2012), which discounts all cash flows exclusively to the fund inception date. Ultimately, selecting a set of compounding dates $\mathcal{T}$ that spans a significant portion of the fund lifecycle appears to decrease finite-sample bias (and to a lesser extent, variance) more effectively than all other measures combined. Appendix A explains the theoretical foundation of this observation, illustrating how our biased estimator (cf. Eq. 10) achieves superior finite-sample properties by exploiting the bias–variance tradeoff.

4 Empirical results for BO and VC data

Following the conclusions from the simulation study, we use the Preqin data from Section 3.1 restricted to North American, USD-denominated funds and form vintage-year portfolios (VYPs) to estimate simple linear SDF models for the Buyout (BO) and Venture Capital (VC) sub-samples individually. This restriction ensures consistency between fund cash flows and the U.S. q-investment factor benchmarks used throughout the analysis. For each model specification, we report asymptotic standard errors (SE_A) from the SHAC framework alongside hv-block cross-validation fold dispersions (SD_{CV}), defined as the standard deviation of parameter estimates across hold-out folds. We analyze BO and VC as two distinct asset classes with fundamentally different risk profiles and report both equal-weighted (EW) and value-weighted (FW) estimates across a grid of compounding Horizons. Throughout this section, t_A denotes the t-ratio based on asymptotic standard errors; SR_{CV} (stabil-

[12]The favorable tradeoff observed here may partly reflect the conditioning on a single realized factor path, which attenuates the TRP covariance terms; see Appendix A for a detailed discussion.

ity ratio) denotes the ratio of the full-sample estimate to the cross-validation fold dispersion, i.e., $\mathrm{SR}_{CV} := \hat{\theta}/\mathrm{SD}_{CV}$[13]. As emphasized in Remark 5, SR_{CV} is a stability diagnostic and should not be interpreted as a test statistic with a known null distribution.

Summary BO data. The Buyout (North America) sample contains 933 distinct funds spreading over 36 vintage years. The vintage year distribution is as follows: 1985 (2), 1986 (3), 1987 (5), 1988 (6), 1990 (2), 1991 (3), 1992 (6), 1993 (5), 1994 (14), 1995 (11), 1996 (13), 1997 (14), 1998 (32), 1999 (27), 2000 (34), 2001 (21), 2002 (14), 2003 (21), 2004 (24), 2005 (44), 2006 (50), 2007 (49), 2008 (45), 2009 (19), 2010 (31), 2011 (25), 2012 (39), 2013 (36), 2014 (44), 2015 (35), 2016 (57), 2017 (35), 2018 (45), 2019 (65), 2020 (34), 2021 (23). The region distribution is as follows: Americas (4), Asia (16), Canada (1), Central and East Europe (2), Europe (6), Japan (1), Midwest (2), North America (461), South America (2), Southeast (1), Southwest (1), US (433), West Europe (1). The strategy distribution is as follows: Buyout (933). The fund currency distribution is as follows: USD (933).

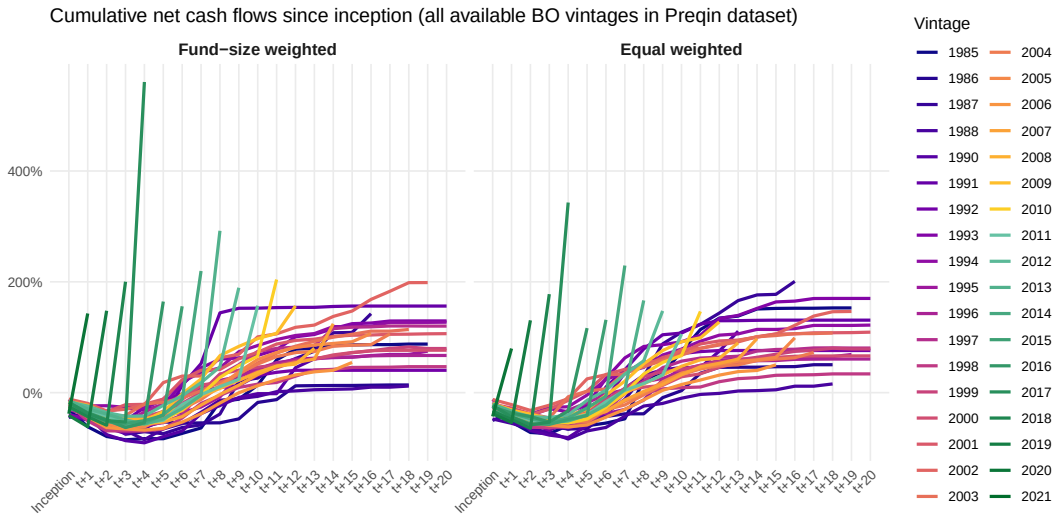

Figure 8: **Cumulative cash flows:** J-curves by vintage year for BO funds in Preqin dataset scaled by fund commitment. To reduce the reliance on large unrealized NAVs, vintage years younger than 2010 are excluded (green lines).

Summary VC data. The Venture Capital (North America) sample contains 968 distinct funds spreading over 38 vintage years. The vintage year distribution is as follows: 1982 (1), 1985 (3), 1986 (4), 1987 (2), 1988 (1), 1989 (1), 1990 (4), 1991 (1), 1992 (7), 1993 (6), 1994 (6), 1995 (8), 1996 (14), 1997 (16), 1998 (28), 1999 (44), 2000 (70), 2001 (45), 2002 (28), 2003 (18), 2004 (31), 2005 (34), 2006 (44), 2007 (48), 2008 (38), 2009 (18), 2010 (18), 2011 (23), 2012 (21), 2013 (32), 2014 (33),

[13]All R code and data are available in an online repository: `https://github.com/quant-unit/Fundwise_SDF/tree/master/r_project`.

2015 (37), 2016 (40), 2017 (50), 2018 (52), 2019 (64), 2020 (45), 2021 (33). The region distribution is as follows: Asia (10), Australasia (1), Canada (1), China (10), Europe (3), India (2), Mexico (1), Midwest (2), North America (267), Northeast (2), Southwest (1), Thailand (1), US (664), West (1). The strategy distribution is as follows: Early Stage (274), Early Stage: Seed (47), Early Stage: Start-up (26), Expansion / Late Stage (127), Venture (General) (472), Venture Debt (22). The fund currency distribution is as follows: USD (968).

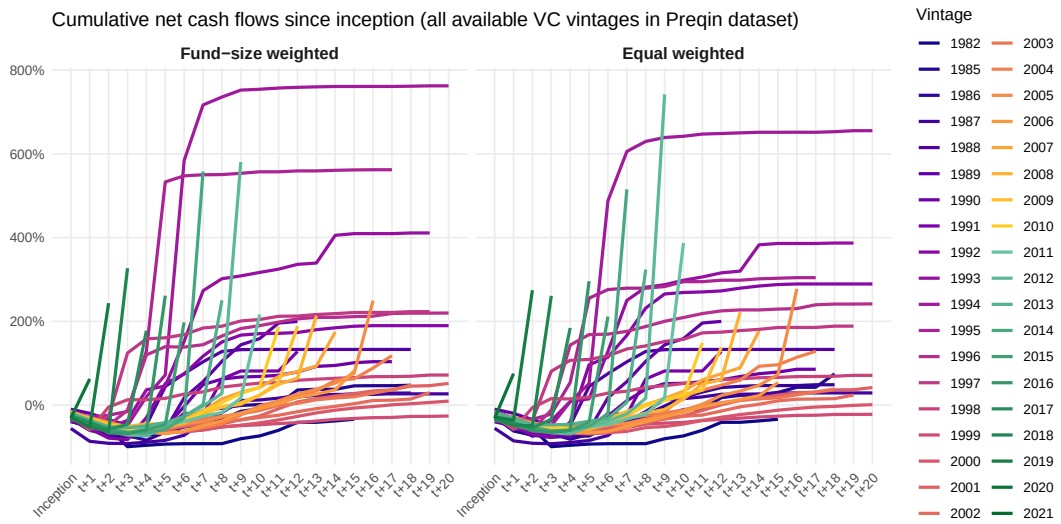

Figure 9: **Cumulative cash flows:** J-curves by vintage year for VC funds in Preqin dataset scaled by fund commitment. To reduce the reliance on large unrealized NAVs, vintage years younger than 2010 are excluded (green lines).

4.1 The impact of unseasoned funds and reported NAVs

Figure 10 presents the vintage-year cutoff sensitivity analysis separately for BO and VC under value weighting (an analogous analysis for aggregate PE is reported in Appendix **??**). The same upward-drift pattern documented for PE is clearly present in both sub-samples: as more recent, unseasoned vintages are included, the estimated market beta increases monotonically. For BO under value weighting, $\hat{\beta}_{\text{MKT}}$ rises from approximately 1.23 with a 2011 vintage cutoff to roughly 2.42 with the full sample (2021); notably, most of the increase is concentrated in vintages after 2016, where $\hat{\beta}_{\text{MKT}}$ jumps sharply from 1.53 (2016 cutoff) to 2.34 (2017 cutoff), reflecting the disproportionate influence of young, unrealized funds. For VC, the drift is similarly dramatic: the FW estimate climbs from approximately 1.57 (2011 cutoff) to 2.33 with the full sample. As with PE, discounting all final NAVs by 50% substantially flattens the upward trend: the BO FW estimate range compresses to 1.09–1.65, and the VC FW range to 1.41–1.54, confirming that the drift is attributable to the outsized mechanical reliance on subjectively reported Net Asset Values for young funds. These findings reinforce the sample restriction to the 2010 maximum vintage year applied throughout the remainder of this section.

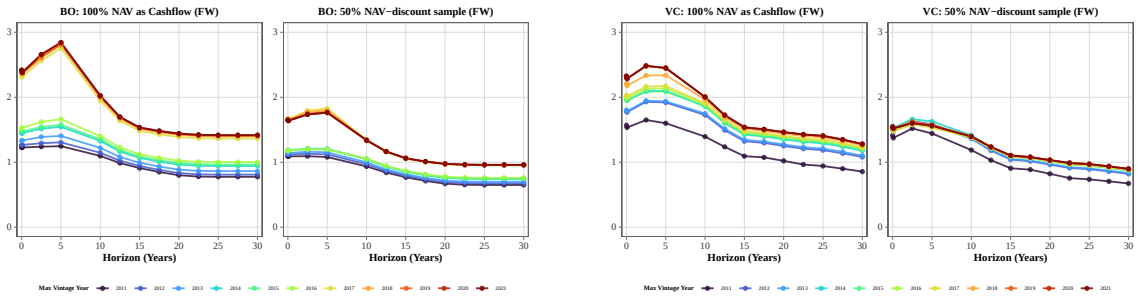

Figure 10: **Sensitivity analysis (MKT factor):** Single-factor model for BO (left) and VC (right) under value weighting, using VYP with different vintage-year cutoffs (asymptotic estimates). For each asset class, the left chart shows the standard baseline (full NAV as final cash flow), while the right chart applies a 50% discount to all final NAVs.

4.2 NAV discount sensitivity analysis

To further quantify the sensitivity of factor loading estimates to terminal Net Asset Value treatment, we conduct a systematic NAV discount analysis on the seasoned sample (maximum vintage year 2010). Figure 11 reports single-factor MKT estimates under five NAV treatments: the baseline (full reported NAV as final cash flow), 25%, 50%, and 75% constant discounts applied to all final NAVs, and a complete NAV write-off (100% discount).

For BO, the estimated market beta decreases monotonically with the NAV discount. At the 0-year Horizon under EW, $\hat{\beta}_{\text{MKT}}$ drops from 1.26 (no discount) to 1.19 (25% discount), 1.12 (50%), 1.03 (75%), and 0.94 (100% write-off); under FW, the corresponding range is 1.20 to 0.98. At the 15-year Horizon, the EW estimate ranges from 0.81 (no discount) to 0.55 (100% write-off), while the FW estimate spans 0.84 to 0.62. The qualitative conclusion is robust across all NAV treatments: the single-factor market model remains economically plausible and stable.

For VC under EW, the pattern is analogous but the sensitivity is more muted in percentage terms due to the higher baseline beta: the 0-year estimate declines from 1.88 (no discount) to 1.84 (25%), 1.80 (50%), 1.77 (75%), and 1.74 (100%); at the 15-year Horizon, from 1.30 (no discount) to 1.09 (100% write-off). Under FW, the sensitivity is more pronounced, particularly at longer Horizons: the 0-year estimate ranges from 1.59 to 1.34, while the 15-year estimate declines from 1.07 to 0.73.

This analysis demonstrates that terminal NAV treatment has a material but bounded effect on estimated factor loadings. The monotonic relationship confirms the intuition that reported NAVs mechanically inflate estimated betas: funds whose final cash flow is a subjective valuation rather than a realized exit price appear more systematically risky because their NAVs partially co-move with public market indices. Importantly, even under the extreme assumption of a complete NAV write-off, the qualitative conclusions of this section remain unchanged: BO maintains a market beta near or slightly below unity, while VC retains a substantially higher systematic exposure. We do not impose a model-based residual-value adjustment, as this would constitute a separate research agenda; instead, the range spanned by

the five NAV treatments provides a transparent envelope of estimation uncertainty attributable to terminal value measurement.

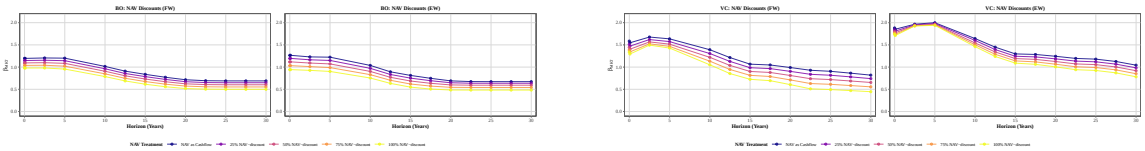

Figure 11: **NAV discount sensitivity (MKT factor):** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010. Each panel shows asymptotic estimates under five NAV treatments: full NAV as final cash flow, 25%, 50%, 75% constant discounts, and 100% NAV write-off.

4.3 Single market-factor models

Tables 5–6 and Figure 12 report the single-factor MKT model for BO and VC across Horizons from 0 to 30 years. As expected from the broader PE results, the market beta declines with the compounding Horizon for both asset classes.

For BO, the estimates are close to their PE counterparts: under EW, $\hat{\beta}_{\text{MKT}}$ ranges from approximately 1.22–1.27 at short Horizons (0–5 years) and decreases to 0.67 at 25–30 years; under FW, the corresponding range is 1.20–1.21 down to 0.69. The divergence between asymptotic and cross-validation diagnostics persists: asymptotic t-ratios (t_A) are moderate at short Horizons (e.g., 1.10 under EW at 0 years) and decline further at longer Horizons, while the cross-validation stability ratios remain high throughout (SR_{CV} ranging from 3.75 to 10.65 under EW and from 2.63 to 8.68 under FW), indicating robust stability across data subsets.

For VC, the market beta is substantially higher across all Horizons, consistent with venture capital's greater sensitivity to systematic risk. Under EW, $\hat{\beta}_{\text{MKT}}$ reaches approximately 1.97–2.00 at 2.5–5 years and converges toward 1.04–1.13 beyond 25 years; under FW, the corresponding range is 1.64–1.68 down to 0.82–0.86. Despite the higher parameter variability reflected in larger asymptotic standard errors (SE_A of 1.2–1.8), the cross-validation stability diagnostics indicate robust estimates (SR_{CV} of 3.59–4.46 under EW; 3.54–7.01 under FW), although somewhat lower than the BO stability ratios due to the more heterogeneous VC fund universe. The converged long-horizon estimates (BO $\approx$ 0.67, VC $\approx$ 1.0–1.1 under EW) are economically plausible: BO funds, which invest in mature companies with stable cash flows, exhibit market exposure comparable to or slightly below that of broad PE, while VC funds, which back early-stage ventures with highly uncertain outcomes, retain a substantially higher beta even at seasoned Horizons.

4.4 Two-factor $\alpha + MKT$ models

Tables 7–8 and Figure 13 present the two-factor MKT + α model for BO and VC. The results show the pathological behavior anticipated by the simulation study:

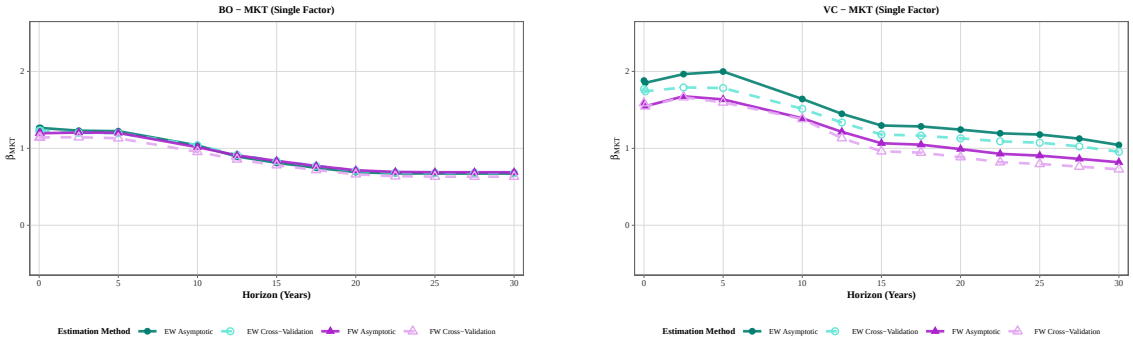

Figure 12: **Empirical (MKT factor):** Single-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

Table 5: Empirical Estimates: BO — Single-Factor (MKT) Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.264	1.154	0.123	1.10	10.00	1.197	1.170	0.139	1.02	8.23
0.1 yr	1.265	1.155	0.116	1.10	10.65	1.196	1.170	0.132	1.02	8.68
2.5 yr	1.229	0.975	0.133	1.26	9.07	1.206	1.175	0.170	1.03	6.73
5 yr	1.223	0.972	0.147	1.26	8.16	1.201	1.173	0.196	1.02	5.78
10 yr	1.038	1.642	0.189	0.63	5.52	1.015	1.806	0.228	0.56	4.19
12.5 yr	0.894	1.660	0.224	0.54	4.08	0.909	1.627	0.247	0.56	3.46
15 yr	0.812	1.511	0.223	0.54	3.75	0.837	1.798	0.246	0.47	3.18
17.5 yr	0.747	1.405	0.206	0.53	3.75	0.771	1.699	0.248	0.45	2.90
20 yr	0.690	2.001	0.163	0.34	4.36	0.715	1.737	0.245	0.41	2.71
22.5 yr	0.674	1.947	0.136	0.35	5.03	0.693	1.687	0.241	0.41	2.65
25 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63
27.5 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63
30 yr	0.672	1.942	0.131	0.35	5.18	0.689	1.678	0.240	0.41	2.63

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

Table 6: Empirical Estimates: VC — Single-Factor (MKT) Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
0 yr	1.881	1.765	0.402	1.07	4.40	1.586	1.645	0.359	0.96	4.42
0.1 yr	1.852	1.756	0.390	1.06	4.46	1.548	1.651	0.335	0.94	4.62
2.5 yr	1.965	1.803	0.451	1.09	3.97	1.677	1.646	0.453	1.02	3.67
5 yr	1.999	1.824	0.493	1.10	3.62	1.636	1.643	0.437	1.00	3.65
10 yr	1.642	1.273	0.364	1.29	4.15	1.391	1.783	0.318	0.78	4.35
12.5 yr	1.449	1.269	0.304	1.14	4.39	1.214	2.005	0.151	0.61	7.49
15 yr	1.297	1.215	0.264	1.07	4.46	1.066	6.793	0.137	0.16	7.01
17.5 yr	1.285	1.216	0.279	1.06	4.17	1.048	6.123	0.148	0.17	6.36
20 yr	1.243	1.220	0.283	1.02	4.00	0.990	4.655	0.159	0.21	5.55
22.5 yr	1.195	1.320	0.272	0.91	4.02	0.929	4.906	0.176	0.19	4.66
25 yr	1.179	1.314	0.272	0.90	3.95	0.905	4.420	0.183	0.20	4.36
27.5 yr	1.126	1.292	0.269	0.87	3.81	0.863	3.749	0.194	0.23	3.93
30 yr	1.043	1.600	0.266	0.65	3.59	0.819	3.220	0.206	0.25	3.54

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.005$.

once an unconstrained alpha intercept is introduced, the estimation destabilizes and the MKT loading collapses.

For BO, the alpha term absorbs a substantial share of the explanatory power: under EW, $\hat{\alpha}$ ranges from 0.003 to 0.006 per month and the MKT coefficient oscillates erratically between 0.41 and 1.01, with no stable convergence across Horizons; under FW, $\hat{\alpha}$ is persistently around 0.005–0.008 per month while the MKT loading collapses to values between 0.10 and 0.32. The asymptotic standard errors are highly erratic, with several entries at or very near zero (indicating near-degenerate regions of the objective function) interspersed with values exceeding 25. Cross-validation fold dispersions for $\hat{\beta}_{\mathrm{MKT}}$ are large (0.09–0.49 under EW; 0.65–0.74 under FW), confirming that the collapsed point estimates are unstable across data subsets.

For VC, the two-factor model exhibits a distinctive and economically puzzling pattern. At short Horizons (0–0.1 years under EW), the alpha term reaches its upper bound ($\hat{\alpha} = 0.020$/month) and the MKT coefficient drops to approximately 0.47, resembling the BO and PE behavior. However, as the Horizon increases beyond 5 years, the MKT loading rises steadily to values exceeding 2.3, while the alpha estimate turns progressively negative, reaching -0.009/month beyond 25 years. This reversal is unique to VC and likely reflects the extreme idiosyncratic risk inherent to early-stage ventures: at long Horizons, the optimizer increasingly attributes the high compounded returns of surviving VC funds to market co-movement rather than alpha, while the negative alpha compensates for selection effects among the funds that underperformed. Under FW, the instability is somewhat attenuated ($\hat{\alpha}$ remains close to zero at short Horizons; $\hat{\beta}_{\mathrm{MKT}} \approx 1.73$), but the overall fragility persists, with asymptotic standard errors exceeding 7–11 across most Horizons.

The BO and VC two-factor results therefore corroborate the conclusion drawn for PE: single-factor MKT models are the most statistically robust specification

for private equity SDF estimation in our sample, given the limited cross-section of vintage-year portfolios.

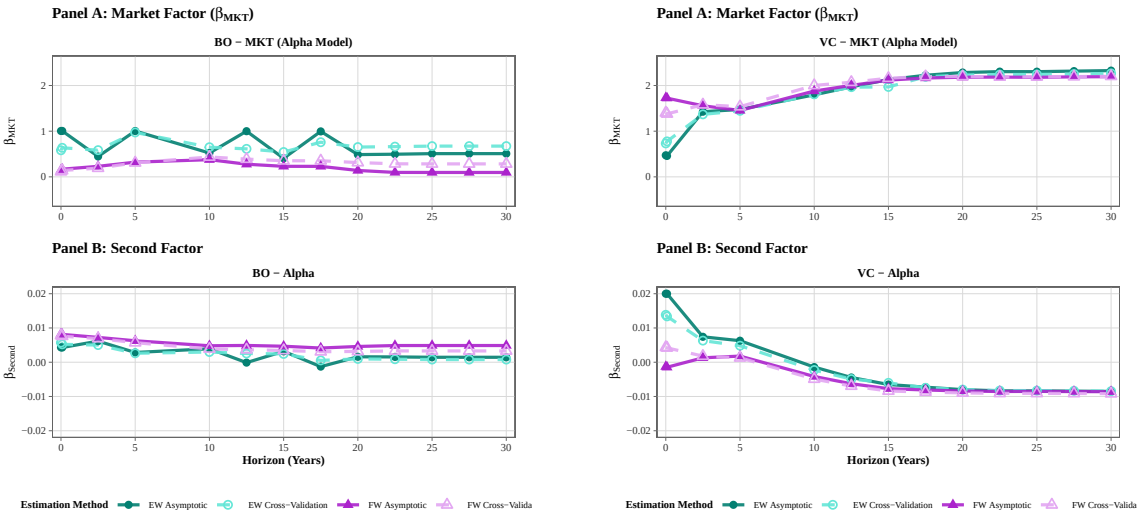

Figure 13: **Empirical ($MKT + \alpha$ model):** Two-factor model for BO (left) and VC (right) using VYP and maximum vintage year 2010.

4.5 Two-factor q^5-factor models

We now pair MKT with each of the four non-market q^5 factors (ME, IA, ROE, EG) in two-factor specifications estimated separately for BO and VC. Because the detailed estimation tables contain extensive Horizon-by-Horizon results, we do not reproduce them here and instead summarize the key patterns visible in Figures 14 and 15; the full tables are available in the accompanying online repository.

Buyout. Figure 14 shows a uniform pattern across the four non-market factors. At short Horizons (0–5 years), the second-factor loadings are positive and economically large: the ME loading reaches approximately 1.4–1.6 under EW (with stability ratios SR_{CV} of 3.0–3.8), the IA loading starts near 1.0 under EW ($\mathrm{SR}_{CV} \approx 1.4$), ROE near 0.6 ($\mathrm{SR}_{CV} \approx 0.9$), and EG near 0.5 ($\mathrm{SR}_{CV} \approx 1.3$). As the Horizon increases, however, all four second-factor loadings decay toward zero and frequently change sign beyond 10–15 years. The ME factor retains a moderate positive loading even at long Horizons under FW ($\hat{\beta}_{\mathrm{ME}} \approx 1.3$, $\mathrm{SR}_{CV} \approx 2.1$), but under EW the loading drops below 1.0 after 17.5 years. The EG, IA, and ROE loadings all fluctuate around zero at intermediate and long Horizons, with cross-validation stability ratios well below conventional thresholds. The MKT loading in these two-factor models remains similar to its single-factor counterpart (ranging from roughly 1.0–1.1 at short Horizons to 0.63–0.76 at long Horizons), showing that a second factor does not materially alter the market risk assessment. The asymptotic standard errors are erratic throughout, often exceeding 10 and occasionally reaching triple-digit values, which highlights the unreliability of asymptotic inference in this setting.

Table 7: Empirical Estimates: BO — MKT + Alpha Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	1.005	1.066	0.489	0.94	1.19	0.147	25.778	0.711	0.01	0.16
0.1 yr	1.005	1.069	0.430	0.94	1.48	0.168	24.142	0.716	0.01	0.19
2.5 yr	0.446	1.379	0.377	0.32	1.55	0.230	6.788	0.739	0.03	0.26
5 yr	1.003	2.078	0.087	0.48	11.22	0.322	6.558	0.707	0.05	0.43
10 yr	0.522	40.420	0.298	0.01	2.18	0.376	3.337	0.652	0.11	0.67
12.5 yr	1.000	3.644	0.386	0.27	1.59	0.277	5.044	0.651	0.05	0.60
15 yr	0.406	0.385	0.357	1.06	1.53	0.231	0.453	0.682	0.51	0.52
17.5 yr	0.995	2.953	0.364	0.34	2.08	0.230	11.801	0.707	0.02	0.50
20 yr	0.488	0.788	0.435	0.62	1.50	0.140	14.154	0.723	0.01	0.43
22.5 yr	0.495	0.779	0.451	0.64	1.47	0.097**	0.035	0.726	2.75	0.39
25 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
27.5 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
30 yr	0.508	0.769	0.447	0.66	1.51	0.096**	0.031	0.726	3.08	0.39
Panel B: Alpha (β_{Alpha})										
0 yr	0.004	0.006	0.004	0.74	1.54	0.008	0.192	0.005	0.04	1.50
0.1 yr	0.004	0.006	0.003	0.73	1.58	0.008	0.179	0.005	0.05	1.46
2.5 yr	0.006	0.014	0.003	0.45	1.83	0.007	0.055	0.005	0.13	1.30
5 yr	0.003	0.013	0.002	0.22	1.45	0.006	0.052	0.005	0.12	1.18
10 yr	0.004	0.363	0.002	0.01	1.27	0.005	0.019	0.004	0.26	0.87
12.5 yr	-0.000	0.124	0.003	-0.00	0.95	0.005	0.036	0.005	0.14	0.81
15 yr	0.003	0.002	0.002	1.88	0.99	0.005	0.003	0.005	1.59	0.73
17.5 yr	-0.001	0.161	0.003	-0.01	0.21	0.004	0.103	0.005	0.04	0.62
20 yr	0.002	0.023	0.003	0.07	0.30	0.005	0.134	0.005	0.03	0.63
22.5 yr	0.002	0.023	0.003	0.07	0.26	0.005***	0.000	0.005	10.88	0.66
25 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66
27.5 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66
30 yr	0.001	0.024	0.003	0.06	0.23	0.005***	0.000	0.005	10.84	0.66

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

Table 8: Empirical Estimates: VC — MKT + Alpha Model

	Equal-Weighted (EW)					Value-Weighted (FW)				
Horizon	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}	$\hat{\beta}$	SE_A	SD_{CV}	t_A	SR_{CV}
Panel A: Market Factor (β_{MKT})										
0 yr	0.473	18.860	0.187	0.03	3.91	1.732	11.028	0.585	0.16	2.40
0.1 yr	0.458	21.158	0.201	0.02	3.86	1.723***	0.237	0.575	7.26	2.40
2.5 yr	1.427	1.806	0.373	0.79	3.67	1.556	9.260	0.341	0.17	4.62
5 yr	1.475	2.296	0.326	0.64	4.44	1.454	8.251	0.323	0.18	4.77
10 yr	1.796	25.246	0.190	0.07	9.49	1.879	17.174	0.237	0.11	8.44
12.5 yr	1.970	4.329	0.205	0.46	9.57	2.002	10.569	0.159	0.19	13.06
15 yr	2.129	6.375	0.441	0.33	4.46	2.119	7.601	0.155	0.28	13.94
17.5 yr	2.224	5.140	0.243	0.43	9.01	2.164	7.186	0.195	0.30	11.27
20 yr	2.284	4.663	0.265	0.49	8.43	2.182	6.896	0.231	0.32	9.53
22.5 yr	2.304	4.669	0.266	0.49	8.41	2.184***	0.316	0.248	6.91	8.81
25 yr	2.303	4.669	0.270	0.49	8.29	2.179***	0.323	0.243	6.75	9.02
27.5 yr	2.315	4.607	0.271	0.50	8.31	2.189***	0.283	0.237	7.73	9.28
30 yr	2.326	4.542	0.268	0.51	8.46	2.193***	0.272	0.234	8.05	9.41
Panel B: Alpha (β_{Alpha})										
0 yr	0.020	0.115	0.009	0.17	1.61	-0.001	0.048	0.010	-0.03	0.45
0.1 yr	0.020	0.130	0.009	0.15	1.46	-0.001	0.001	0.010	-1.84	0.43
2.5 yr	0.007	0.009	0.005	0.81	1.35	0.001	0.043	0.003	0.03	0.60
5 yr	0.006	0.012	0.004	0.54	1.14	0.002	0.039	0.003	0.05	0.37
10 yr	-0.001	0.119	0.002	-0.01	-1.30	-0.004	0.095	0.001	-0.04	-4.27
12.5 yr	-0.004	0.006	0.001	-0.78	-5.86	-0.006	0.059	0.001	-0.11	-7.52
15 yr	-0.007	0.036	0.002	-0.18	-2.99	-0.008	0.048	0.001	-0.16	-7.27
17.5 yr	-0.007	0.032	0.001	-0.23	-9.95	-0.008	0.046	0.001	-0.18	-6.11
20 yr	-0.008	0.030	0.001	-0.27	-8.85	-0.008	0.045	0.002	-0.19	-5.30
22.5 yr	-0.008	0.030	0.001	-0.28	-8.18	-0.009***	0.001	0.002	-9.67	-4.97
25 yr	-0.008	0.030	0.001	-0.28	-8.14	-0.009***	0.001	0.002	-9.61	-5.07
27.5 yr	-0.008	0.029	0.001	-0.29	-8.45	-0.009***	0.001	0.002	-9.74	-5.22
30 yr	-0.009	0.029	0.001	-0.29	-8.66	-0.009***	0.001	0.002	-9.81	-5.36

Notes: $\hat{\beta}$ = parameter estimate; SE_A = asymptotic standard error; SD_{CV} = cross-validation fold dispersion (standard deviation across hold-out folds); t_A = asymptotic t-ratio; SR_{CV} = stability ratio (estimate / fold dispersion). Significance markers based on t_A: $^{*}\,p < 0.05$, $^{**}\,p < 0.01$, $^{***}\,p < 0.005$.

Venture Capital. Figure 15 displays qualitatively analogous but more extreme patterns, consistent with the higher idiosyncratic risk inherent to early-stage ventures. At short Horizons, the second-factor loadings are strikingly large: the IA loading reaches 5.1 under EW at the 0-year Horizon ($\mathrm{SR}_{CV} \approx 1.4$) and 4.2 under FW, the ROE loading starts at 1.5 (EW) and 1.0 (FW), the EG loading at approximately 3.5 (EW) and 6.6 (FW), while the ME factor is the exception with a subdued loading near zero or slightly positive (0.3 under EW, -0.03 under FW). All four second-factor loadings undergo dramatic sign reversals as the Horizon extends. The IA loading, for instance, turns negative beyond 10 years and eventually reaches -1.5 to -1.9 at long Horizons with large stability ratios (SR_{CV} exceeding -12). Similarly, EG and ROE produce large negative loadings at long Horizons ($\hat{\beta}_{\mathrm{EG}} \approx -0.9$, $\hat{\beta}_{\mathrm{ROE}} \approx -1.2$ under EW at 30 years). These sign reversals echo the behavior documented in the simulation study (Section 3.3), where two-factor models exhibited severe attenuation and sign changes in the second-factor loading across Horizons. The MKT loading in the VC two-factor models varies substantially across specifications: in the IA model it stabilizes around 1.4–1.5 at intermediate Horizons, while in the ROE and EG models it ranges from 0.4 to 2.0 depending on the Horizon and weighting scheme.

Summary. The q^5-factor results for BO and VC confirm the fragility of multi-factor SDF estimation documented in the preceding MKT + α analysis. While some second-factor loadings attain statistical significance at isolated Horizons (particularly ME for BO and short-Horizon IA for VC), none of the four non-market factors produces a stable, economically interpretable loading across the full Horizon spectrum. The pervasive sign reversals and the extreme sensitivity of point estimates to the compounding Horizon indicate that these results are driven by finite-sample noise rather than genuine systematic risk exposures. We conclude that the q^5-factor two-factor models do not improve upon the single-factor MKT benchmark for either asset class.

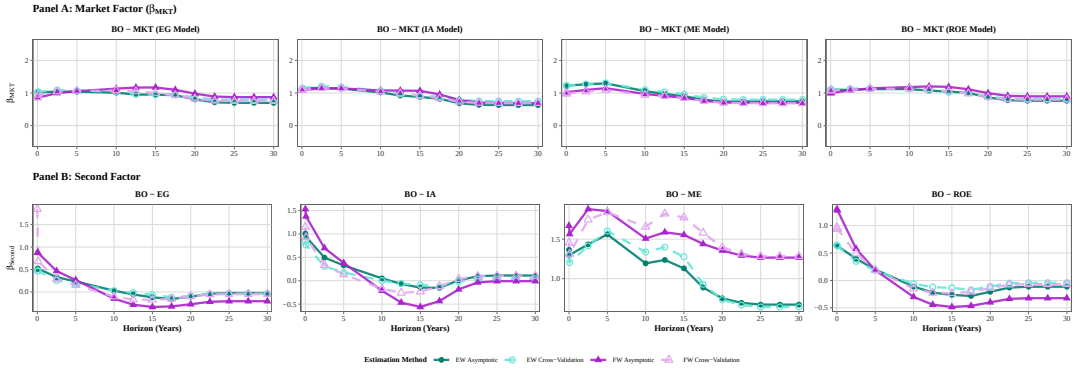

Figure 14: **Empirical results (q^5-factors):** Two-factor model for BO funds using VYP and maximum vintage year 2010.

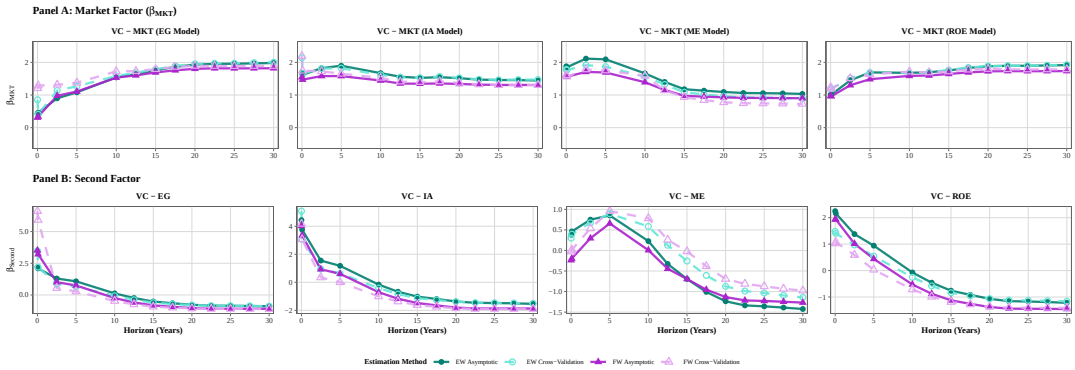

Figure 15: **Empirical results (q^5-factors):** Two-factor model for VC funds using VYP and maximum vintage year 2010.

4.6 Interpretation

The empirical results for Buyout and Venture Capital funds demonstrate consistent model reliability across both asset classes and weighting schemes.

Single-factor MKT models are the most robust specification. The single-factor market model produces economically plausible and statistically significant estimates for both BO and VC. For Buyout, the market beta ranges from approximately 1.27 at the shortest Horizon to 0.67 at long Horizons under equal weighting, with cross-validation stability ratios consistently exceeding 3.7. For Venture Capital, the corresponding range is 1.88–2.00 down to 1.04–1.13, reflecting the substantially higher systematic risk exposure of early-stage investments. The cross-validation stability diagnostics are strong and consistent across the entire Horizon spectrum, even as asymptotic standard errors fluctuate erratically. The divergence between asymptotic inference and cross-validation diagnostics mirrors the simulation finding that, with the limited cross-section of vintage-year portfolios, the asymptotic approximation is unreliable and overly conservative.

Multi-factor models are fragile. Both the MKT + α and the q^5-factor two-factor models collapse as soon as a second parameter is introduced. In the MKT + α specification, the alpha intercept absorbs explanatory power from MKT, producing collapsed or negative market betas and erratic standard errors. The q^5-factor models exhibit a pervasive pattern of sign reversals: second-factor loadings are consistently overestimated at short Horizons and attenuate toward or beyond zero at long Horizons, exactly as predicted by the simulation study in Section 3.3. This fragility is not confined to a single factor or asset class; it appears across all eight specifications (four factors $\times$ two asset classes), ruling out the chance that an individual q^5 factor might robustly augment the market model.

BO–VC decomposition. Analyzing Buyout and Venture Capital separately provides a more informative test of model stability than reporting aggregate PE

results alone. BO and VC represent fundamentally different investment strategies: Buyout targets mature companies with leveraged balance sheets and stable cash flows, while Venture Capital finances early-stage firms with highly uncertain, option-like payoff profiles. Despite these differences, both sub-samples exhibit the same diagnostic pattern: robust single-factor significance, alpha-induced destabilization, and second-factor fragility. The fact that this pattern appears across two asset classes with different risk characteristics suggests that the multi-factor instability is a finite-sample identification problem, not an artifact of pooling heterogeneous fund types.

Reconciliation with the simulation study. The empirical results mirror the simulation predictions from Section 3.3. First, the MSE-minimizing Horizon at which the single-factor estimates stabilize (approximately 15–20 years) aligns with the 12.5–15-year optimum identified in simulation. Second, the sign reversals and large estimation variance observed in the empirical q^5-factor models replicate the pathological behavior documented in the two-factor simulation experiments, where the second-factor loading underwent severe attenuation and occasionally changed sign across Horizons. Third, the divergence between asymptotic inference and cross-validation stability diagnostics in the data is consistent with the simulation finding that asymptotic standard errors systematically overstate true parameter uncertainty in small samples. These results show that the finite-sample challenges identified by our simulation study appear directly in empirical estimation with real-world private market data.

Implications for empirical practice. Our findings yield a clear recommendation for applied researchers and practitioners. In our sample, semiparametric SDF estimation for private equity cash flows is substantially more reliable when restricted to single-factor market models estimated on seasoned, vintage-year portfolios at an extended compounding Horizon of approximately 15 years. Multi-factor models and alpha intercepts, while theoretically appealing, cannot be reliably estimated with the cross-sectional sample sizes currently available in private equity databases. Only a substantial expansion of the available data, either through longer time series of fully seasoned funds or through larger cross-sections within each vintage year, would provide the statistical power necessary to disentangle multiple systematic risk exposures from finite-sample noise.

5 Conclusion

Our generalized Least-Mean-Distance framework provides a flexible theoretical foundation for estimating Stochastic Discount Factor (SDF) models using non-traded cash flows. However, our Monte Carlo simulations reveal that semiparametric estimators in this setting face severe finite-sample challenges. Specifically, we demonstrate that the foundational Driessen et al. (2012) estimator exhibits substantially greater bias and variance than previously recognized, particularly concerning the estimation of abnormal performance (α). Our framework shows that this finite-sample

bias can be dramatically mitigated by averaging pricing errors across multiple future dates spanning the fund lifecycle, while estimator variance is most effectively controlled by increasing the number of cross-sectional units (n).

In the data-sparse private equity domain, where estimation relies on a limited cross-section of vintage-year portfolios, asymptotic approximations prove unreliable. We therefore strongly advocate for the use of robust resampling or cross-validation stability diagnostics tailored to overlapping cash flow dependencies. While relying on individual fund data yields smaller asymptotic standard errors, our simulations indicate that this approach exacerbates finite-sample bias and estimation noise, rendering it an unsuitable remedy. Given that virtually all multi-factor models in our empirical analysis fail to yield consistently stable parameter estimates under either asymptotic inference or cross-validation stability diagnostics, and this fragility is replicated independently for both Buyout and Venture Capital funds despite their markedly different risk profiles, we conclude that, in our sample, single-factor MKT specifications are substantially more stable than multi-factor alternatives and should be preferred (without α terms) until significantly broader cross-sections of private equity data become available. Researchers must exercise caution even then, ensuring that data-driven model selection (e.g., via hv-block cross-validation) does not yield economically implausible outcomes.

Expanding the SDF framework to robustly accommodate multiple risk factors requires imposing additional economic or statistical structure. Future research could further explore parametric assumptions for the cash flow generating process (Ang et al., 2018) or extract intra-period information from reported Net Asset Values (Gredil et al., 2020; Brown et al., 2021). Methodologically, addressing the high estimator variance demands statistical techniques suited for sparse, high-dimensional settings. Approaches such as LASSO, ridge regression, or model averaging to regularize non-market coefficients, as well as ensemble learning methods that combine multiple weak learners, present promising paths forward.

Another vital avenue for future work involves explicitly modeling complex fee structures and carried interest within the data generating process. This would enable the estimation of gross-versus-net-of-fee SDFs, providing sharper evaluation models for both General Partners (GPs) and Limited Partners (LPs). Such structural refinements would naturally improve parameter accuracy and allow for a more rigorous analytical characterization of finite-sample bias. Additionally, carefully integrating intermediate NAVs as proxies for unobservable true valuations could help alleviate the timing risk premium bias inherent to terminal value concepts, though this requires downweighting actual cash distributions in favor of subjective accounting metrics.

Finally, we highlight the critical role of the compounding Horizon in empirical asset pricing for private markets. Our simulation study demonstrates that utilizing a future value framework, i.e., averaging pricing errors over a broad set of discounting dates $\mathcal{T}$, yields vastly superior finite-sample properties compared to traditional net present value approaches. The bias is minimized when the set of discounting dates corresponds approximately to the typical fund lifecycle. Developing a formal theoretical or empirical mechanism to endogenously determine the optimal configuration of $\mathcal{T}$ for a given fund sample and macroeconomic environment remains an

important objective for future research.

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A Bias–Variance Tradeoff of $\mathcal{T}$-Averaging

This appendix shows that $\mathcal{T}$-averaging introduces a small asymptotic bias via the Timing Risk Premium but achieves a net reduction in finite-sample estimation error by lowering the variance of the pricing-error moments. We formalize this claim and reconcile it with the simulation results from Section 3.3.

A.1 Pseudo-true parameter

For a given maximum compounding Horizon H, we define the *horizon-specific pseudo-true parameter* as the population minimizer of the horizon-averaged objective:

$$\theta_H^* \;:=\; \arg\min_{\theta}\; Q_H(\theta) \;=\; \arg\min_{\theta}\; \mathbb{E}\Big[\bar{\epsilon}_i^{(H)}(\theta)^2\Big] \tag{25}$$

where $\bar{\epsilon}_i^{(H)}(\theta)$ is the average pricing error computed over the set of discounting dates $\mathcal{T}_H = \{1, \ldots, 12 \cdot H\}$ (cf. Remark 6).

Under correct model specification and Assumption 2, the NPV-only estimand coincides with the structural parameter:

$$\theta_0^* = \theta_0$$

because the population moment condition $\mathbb{E}[\bar{\epsilon}_i^{(0)}(\theta_0)] = 0$ is satisfied exactly (Section 2.1, Case A). For $H > 0$, the Timing Risk Premium introduces a non-zero population moment (Equation 29), so that in general

$$\theta_H^* \neq \theta_0$$

The wedge $\|\theta_H^* - \theta_0\|$ is governed by the magnitude of the TRP covariance terms, which are bounded by the Hansen–Jagannathan bound and diluted by the $1/|\mathcal{T}_i|$ averaging (see Section A.5 below). Consequently, the population-level bias from horizon averaging is small relative to the dramatic finite-sample estimation error reduction documented in our simulation study (Section 3.3).

A.2 Decomposition of $\mathbb{E}[\bar{\epsilon}_i(\theta_0)]$

Starting from the average pricing error (Equation 17) and the deal-level decomposition (Equation 3), we can write

$$\bar{\epsilon}_i(\theta_0) = \frac{1}{|\mathcal{T}_i|} \sum_{\tau \in \mathcal{T}_i} \epsilon_{\tau,i}(\theta_0) = \frac{1}{|\mathcal{T}_i|} \sum_{\tau \in \mathcal{T}_i} \sum_{j=1}^{J} \frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\, \delta_{i,j} \tag{26}$$

Taking the unconditional expectation yields

$$\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = \frac{1}{|\mathcal{T}_i|} \sum_{\tau \in \mathcal{T}_i} \sum_{j=1}^{J} \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\, \delta_{i,j}\right] \tag{27}$$

Because we now compute the unconditional expectation rather than conditioning on $\mathcal{F}_\tau$, we can decompose each (τ, j) term via the standard covariance identity $\mathbb{E}[XY] = \mathrm{Cov}(X, Y) + \mathbb{E}[X]\mathbb{E}[Y]$:

$$\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\,\delta_{i,j}\right] = \underbrace{\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau},\,\delta_{i,j}\right)}_{\mathrm{TRP}_{\tau,j}} + \underbrace{\mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\right]}_{>0} \cdot \underbrace{\mathbb{E}[\delta_{i,j}]}_{=0} \tag{28}$$

Hence the asymptotic bias of the average pricing error is entirely determined by the averaged Timing Risk Premium:

$$\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = \frac{1}{|\mathcal{T}_i|}\sum_{\tau\in\mathcal{T}_i}\sum_{j=1}^{J}\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau},\,\delta_{i,j}\right) \tag{29}$$

The key observation is that this covariance is exactly zero when $\tau \le t_{i,j}^{\mathrm{Inv}}$ (Case A from Section 2.1). When $\tau \le t_{i,j}^{\mathrm{Inv}}$, iterated expectations through $\mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}$ give

$$\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau},\,\delta_{i,j}\right) = \mathbb{E}\left[\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau}\,\underbrace{\mathbb{E}\left[\delta_{i,j}\mid\mathcal{F}_{t_{i,j}^{\mathrm{Inv}}}\right]}_{=0\text{ by Eq. 2}}\right] = 0 \tag{30}$$

Therefore the bias decomposes as

$$\mathbb{E}[\bar{\epsilon}_i(\theta_0)] = \frac{1}{|\mathcal{T}_i|}\left[\underbrace{\sum_{\substack{\tau\in\mathcal{T}_i,\,j\\ \tau\le t_{i,j}^{\mathrm{Inv}}}} 0}_{\text{NPV terms}^{14}} + \underbrace{\sum_{\substack{\tau\in\mathcal{T}_i,\,j\\ \tau> t_{i,j}^{\mathrm{Inv}}}}\mathrm{Cov}\left(\frac{\Psi_{t_{i,j}^{\mathrm{Inv}}}}{\Psi_\tau},\,\delta_{i,j}\right)}_{\text{NFV terms (Timing Risk Premium)}}\right] \tag{31}$$

Remark (misspecification). Under model misspecification, $\mathbb{E}[\delta_{i,j}] \neq 0$ and the second term in Equation 28 no longer vanishes. This introduces an additional bias component proportional to $\mathbb{E}[\Psi_{t_{i,j}^{\mathrm{Inv}}}/\Psi_\tau]\cdot\mathbb{E}[\delta_{i,j}]$; while the $1/|\mathcal{T}_i|$ denominator dilutes each additional term, the bias grows with $|\mathcal{T}_i|$ whenever the added NFV terms have non-vanishing $\mathbb{E}[\Psi_{t_{i,j}^{\mathrm{Inv}}}/\Psi_\tau]$, so that the cumulative sum of misspecification terms outpaces the averaging effect.

A.3 Reconciliation with the simulation study

If this asymptotic bias were the only effect, the NPV-only approach of Driessen et al. (2012) with $\mathcal{T}_i = \{\min_j t_{i,j}^{\mathrm{Inv}}\}$ would exhibit the smallest bias, and adding NFV terms would always increase it. However, the simulation study in Section 3.3 shows the opposite: averaging over larger $\mathcal{T}_i$ *reduces* total bias. The resolution involves two competing effects.

[14] The sums over j in both the NPV and NFV branches run over all J deals, but each (τ, j) pair contributes to exactly one branch depending on whether $\tau \le t_{i,j}^{\mathrm{Inv}}$ or $\tau > t_{i,j}^{\mathrm{Inv}}$.

Effect 1: Finite-sample bias from nonlinear estimation We write the nonlinear least-squares objective in its maximization form $Q_n(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\bar{\epsilon}_i(\theta)^2$, so that $\hat{\theta} = \arg\max_\theta Q_n(\theta)$ and the Hessian $H = \nabla^2 Q_n(\theta_0)$ is negative semi-definite at the optimum.[15] The finite-sample bias of $\hat{\theta}$ is approximately (standard Nagar (1959)-type expansion for M-estimators):

$$\mathbb{E}[\hat{\theta} - \theta_0] \approx -\frac{1}{n} H^{-1} \cdot b(\mathrm{Var}[\bar{\epsilon}_i], \text{curvature}) \tag{32}$$

where $H = \nabla^2 Q_n(\theta_0)$ denotes the Hessian of the objective function. The term curvature refers to the second-order derivatives $\nabla^2_\theta \bar{\epsilon}_i(\theta_0)$ of the moment conditions with respect to the parameter vector. These derivatives measure how nonlinear the estimating equations are in θ: if the pricing-error moments were perfectly linear in θ, the curvature would vanish and no finite-sample bias would arise. The correction term $b(\cdot)$ is the Nagar-type bias correction from Rilstone et al. (1996) and Newey and Smith (2004); it captures the interaction between the variance of the moment conditions $\mathrm{Var}[\bar{\epsilon}_i(\theta_0)]$ and the curvature of the objective. Specifically, b is proportional to $\mathrm{Var}[\bar{\epsilon}_i(\theta_0)]$: the noisier the moments, the more the nonlinear optimizer is "pulled away" from the true θ_0 in finite samples, because the concavity/convexity of Q_n causes asymmetric responses to positive and negative moment deviations. When each fund provides only one noisy moment (NPV-only), this variance is high, resulting in a substantial finite-sample bias.

Effect 2: Variance reduction through $\mathcal{T}$-averaging When averaging over $|\mathcal{T}_i|$ discounting dates, the variance of the average pricing error is

$$\mathrm{Var}[\bar{\epsilon}_i] = \frac{1}{|\mathcal{T}_i|^2} \sum_{\tau,\tau' \in \mathcal{T}_i} \mathrm{Cov}\big(\epsilon_{\tau,i},\, \epsilon_{\tau',i}\big) \tag{33}$$

The pricing errors $\epsilon_{\tau,i}$ at different τ are positively correlated (they share the same deal cash flows and factor realizations), but they are not perfectly correlated because different τ's weight the SDF path differently, so that exploiting these additional conditional moment restrictions improves efficiency (Newey, 1993). Hence

$$\mathrm{Var}[\bar{\epsilon}_i] < \mathrm{Var}[\epsilon_{\tau_0,i}] \quad \text{for } |\mathcal{T}_i| > 1$$

and this variance reduction directly decreases the finite-sample bias from the nonlinear optimization (cf. Equation 32).

The net effect The total bias in small samples decomposes approximately as

$$\underbrace{\text{Finite-sample bias}}_{\propto \mathrm{Var}[\bar{\epsilon}_i]/n} + \underbrace{\text{Asymptotic TRP bias}}_{\text{Eq. 31}}$$

As $|\mathcal{T}_i|$ increases from one:

- The finite-sample bias *decreases* because $\mathrm{Var}[\bar{\epsilon}_i]$ drops (cf. Equation 32).

[15] Equivalently, one may minimize $\frac{1}{n}\sum_i \bar{\epsilon}_i^2$; the sign change cancels in the bias expression below.

- The TRP bias *increases* as more NFV terms enter, but is tempered by two factors:
  - only (τ, j) pairs with $\tau > t_{i,j}^{\text{Inv}}$ contribute non-zero TRP terms,
  - the $1/|\mathcal{T}_i|$ denominator dilutes the accumulating terms.

The simulation study finds the sweet spot where these two effects balance: at $|\mathcal{T}_i|$ corresponding to approximately the fund lifetime (180 months), the variance-reduction benefit dominates the TRP cost. Proposition 5 below formalizes this tradeoff and establishes the existence of an interior MSE-minimizing horizon H^*.

A.4 Formal MSE decomposition

The qualitative tradeoff described above can be formalized into an explicit Mean Squared Error (MSE) decomposition that separates the population-level targeting error (from the Timing Risk Premium) and the finite-sample estimation error (from the nonlinear objective). Let θ_0 denote the structural parameter of interest (e.g., the true factor loadings). The standard MSE around this structural parameter decomposes into squared bias and variance:

$$\text{MSE}(\hat{\theta}_H, \theta_0) \;:=\; \mathbb{E}\big[\|\hat{\theta}_H - \theta_0\|^2\big] \;=\; \|\mathbb{E}[\hat{\theta}_H] - \theta_0\|^2 + \mathbb{E}\big[\|\hat{\theta}_H - \mathbb{E}[\hat{\theta}_H]\|^2\big] \tag{34}$$

To understand how this behaves under horizon averaging, we expand this around the horizon-specific pseudo-true parameter θ_H^*.

Proposition 5 (MSE decomposition of the horizon-averaged estimator)**.** *Under the regularity conditions for the finite-sample bias expansion of nonlinear M-estimators (Rilstone et al., 1996; Bao and Ullah, 2007), the mean squared error of $\hat{\theta}_H$ around the structural parameter θ_0 decomposes to leading order as:*

$$\text{MSE}(\hat{\theta}_H, \theta_0) \;\approx\; \underbrace{B_{pop}(H)^2}_{\textit{TRP targeting error}} \;+\; \underbrace{B_{fs}(H)^2}_{\textit{Finite-sample bias}} \;+\; \underbrace{\mathcal{V}(H)}_{\textit{Estimator variance}} \tag{35}$$

where

$$\begin{aligned} B_{pop}(H)^2 &:= \|\theta_H^* - \theta_0\|^2 \\ B_{fs}(H)^2 &:= \|\mathbb{E}[\hat{\theta}_H] - \theta_H^*\|^2 \;\propto\; \text{Var}[\bar{\epsilon}_i^{(H)}(\theta_H^*)]/n \\ \mathcal{V}(H) &:= \text{tr}(\Sigma_H/n) \end{aligned}$$

with Σ_H denoting the asymptotic sandwich covariance matrix evaluated at θ_H^*.

The cross-term $2(\theta_H^* - \theta_0)^\top \mathbb{E}[\hat{\theta}_H - \theta_H^*]$ is absorbed into the leading-order factorization because both bias terms decay with sample size n and horizon H, respectively.

Corollary 1 (Monotonicity in H)**.** *Under the conditions of Proposition 5, the components of the MSE exhibit the following monotonicity properties with respect to the maximum compounding horizon H:*

(i) $B_{pop}(H)$ *is non-decreasing in* H *with* $B_{pop}(0) = 0$*. Each additional NFV term* $\tau > t_{i,j}^{\text{Inv}}$ *contributes a non-negative TRP increment to the population moment (Equation 31), while the* $1/|\mathcal{T}_i|$ *denominator ensures that* $B_{pop}(H)$ *grows sublinearly.*

(ii) $B_{fs}(H)$ *is non-increasing in* H*. Because the pricing errors* $\epsilon_{\tau,i}$ *are imperfectly correlated across* τ*, the variance of the average moment* $\text{Var}[\bar{\epsilon}_i^{(H)}]$ *decreases strictly with* $|\mathcal{T}_i|$*, which in turn monotonically reduces the finite-sample bias induced by the nonlinear objective.*

(iii) $\mathcal{V}(H)$ *is non-increasing in* H*. Averaging across multiple compounding dates corresponds to combining imperfectly correlated moment conditions, which weakly reduces the asymptotic variance of* $\hat{\theta}_H$ *around* θ_H^**.*

Consequently, provided the TRP terms are not identically zero, the total $\text{MSE}(H)$ *admits an interior minimum at some optimal horizon* $H^* \in (0, \infty)$*.*

Remark 8 (Magnitude of the population bias)**.** *While* $B_{pop}(H)$ *grows with* H*, its magnitude is tightly disciplined by fundamental asset pricing bounds. For each deal* j *and discounting date* $\tau > t_{i,j}^{\text{Inv}}$*, the Cauchy–Schwarz inequality applied to the Hansen and Jagannathan (1991) framework bounds the covariance:*

$$\left|\text{Cov}\Big(\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau},\, \delta_{i,j}\Big)\right| \leq \sigma\Big(\frac{\Psi_{t_{i,j}^{\text{Inv}}}}{\Psi_\tau}\Big) \cdot \sigma(\delta_{i,j})$$

The standard deviation of the SDF ratio is bounded by the maximum Sharpe ratio over the $(t_{i,j}^{\text{Inv}}, \tau)$ *horizon times* $\mathbb{E}[\Psi_{t_{i,j}^{\text{Inv}}}/\Psi_\tau]$*. Furthermore, the* $1/|\mathcal{T}_i|$ *averaging dilutes these terms: for a fund with roughly* $12H$ *discounting dates, the cumulative TRP contribution enters* $B_{pop}(H)$ *with a* $1/(12H)$ *scaling factor. Under our baseline simulation data generating process, the resulting TRP population bias* $B_{pop}(H^*)$ *is an order of magnitude smaller than the uncorrected finite-sample bias* $B_{fs}(0)$ *at the zero horizon.*

Remark 9 (DGP dependence of H^*)**.** *The exact location of the interior minimum* H^* *depends sensitively on the underlying data generating process: (1) higher idiosyncratic volatility* σ *increases* B_{fs}*, shifting* H^* *upward; (2) higher factor volatility and risk premiums increase SDF variability and thus* B_{pop}*, shifting* H^* *downward; (3) larger cross-sectional sample sizes* n *shrink* B_{fs}*, shifting* H^* *downward as the relative benefit of horizon averaging diminishes. The simulation results in Section 3.3 empirically confirm this tradeoff and indicate that* $H^* \approx 12$–15 *years under our baseline calibration.*

A.5 Interpretation

The $\mathcal{T}$-averaging mechanism can be understood through three complementary perspectives from the econometrics and asset pricing literature.

Variance reduction via moment combination. In the GMM framework of Hansen (1982), efficiency gains arise from combining multiple moment conditions.

Each discounting date $\tau \in \mathcal{T}_i$ contributes a separate pricing-error moment $\epsilon_{\tau,i}(\theta)$, and averaging over $\mathcal{T}_i$ is formally equivalent to combining $|\mathcal{T}_i|$ such moments into a single summary condition. Since these moments share the same underlying deal cash flows and factor paths but weight the SDF trajectory differently, they are positively yet imperfectly correlated. The resulting reduction in $\mathrm{Var}[\bar{\epsilon}_i]$ is the classical Hansen (1982) effect of exploiting additional moment information, which in turn decreases the finite-sample bias of the nonlinear estimator (cf. Equation 32).

The TRP as a bounded risk-premium term. The Timing Risk Premium in Equation 29 is a covariance between a ratio of stochastic discount factors and a deal-level pricing error. Within the Hansen and Jagannathan (1991) framework, the magnitude of such covariances is bounded by the product of the SDF's variability and the payoff's standard deviation. Under correct model specification, the TRP terms are therefore non-zero (i.e., they reflect genuine systematic risk) but their size is disciplined by the Hansen–Jagannathan bound. Consequently, the TRP bias introduced by NFV terms ($\tau > t_{i,j}^{\mathrm{Inv}}$) grows only moderately with $|\mathcal{T}_i|$ and is further diluted by the $1/|\mathcal{T}_i|$ denominator.

Analogy to model averaging. The structure of $\mathcal{T}$-averaging closely mirrors the jackknife model averaging framework of Hansen and Racine (2012), where combining multiple (possibly misspecified) estimators achieves a bias–variance tradeoff: averaging reduces variance at the expense of a small bias from the inclusion of imperfect components. In our setting, the "models" being averaged are not separate econometric specifications but rather the individual pricing-error moment conditions evaluated at different discounting dates τ. Each individual moment is unbiased for the NPV terms but may carry TRP bias for the NFV terms; analogous to the weak misspecification in the Hansen and Racine (2012) setup. Their key insight applies directly: the variance reduction from averaging dominates the bias cost as long as the individual components are not too severely biased, which is precisely what the Hansen and Jagannathan (1991) bound guarantees in our context.

Summary. The $\mathcal{T}$-averaging mechanism does not make the TRP wash out in the sense that each individual TRP term goes to zero. Rather, it achieves a bias–variance tradeoff where the TRP bias introduced by NFV terms is more than compensated by the variance reduction that cures the finite-sample bias from the nonlinear estimation. The fact that the optimal $|\mathcal{T}_i|$ depends on the SDF model specification is consistent with this explanation: a more complex or misspecified model not only increases the TRP covariance terms but also activates the additional bias channel identified in the Remark on misspecification above, shifting the optimal tradeoff point. Finally, the favorable tradeoff observed in our simulation study may partly reflect the conditioning on a single realized factor path, which attenuates the TRP covariance terms relative to their unconditional magnitude. Across different macro regimes, the TRP cost could be larger and the optimal $|\mathcal{T}_i|$ smaller.

B Comparison to Driessen et al. (2012)

B.1 Comparison to simulation results

(Driessen et al., 2012, Table 1) report Monte Carlo results for $n = 50$ funds per vintage-year portfolio with an annualized idiosyncratic volatility of 108% ($\sigma_{\text{monthly}} = 0.3$). In our base case (S1), we use only $n = 20$ funds per vintage, which is more realistic for older vintages and smaller private capital asset classes beyond Buyout and Venture Capital. Scenarios S2–S9 in Table 9 systematically vary the simulation design to identify which features drive the remaining discrepancy between our results and the Driessen et al. (2012) benchmark.

Several structural differences between the two simulation designs should be noted. First, Driessen et al. (2012) use a shifted lognormal distribution for the idiosyncratic error term (cf. their online appendix, Equation 9), which ensures that the error addend has a finite lower bound (approximately -1) and avoids the nonlinear distortions that arise when normally distributed errors occasionally produce extreme negative total returns. Second, they employ a "full Monte Carlo" approach and simulate the total market return R_{MKT} from a shifted lognormal distribution calibrated to S&P 500 moments (1980–2003), rather than conditioning on the historical path. Third, their deal investment timing follows a deterministic schedule (three deals per year, evenly spaced across the five-year investment period), rather than random uniform draws.

Interpretation of results. Under our original design with normal errors and $\sigma = 30\%$ (S3), the market beta is severely attenuated ($\hat{\beta} = 0.62$) and the spurious alpha (0.68%/month) is an order of magnitude larger than the benchmark (0.05%). Switching to the shifted lognormal error distribution (S4) modestly improves the beta estimate to 0.71 and reduces the alpha bias to 0.57%. Adding simulated market returns (S5) yields no further improvement ($\hat{\beta} = 0.66$), and deterministic investment timing (S7) produces a similarly attenuated beta of 0.70 without substantially reducing alpha variance.

The most effective modification is shortening the maximum deal holding period from 10 years to 5 years (S8), which significantly improves the beta estimate ($\hat{\beta} = 0.91$) and reduces the alpha bias to 0.30% without requiring lower idiosyncratic volatility. Combining this holding-period restriction with deterministic investment timing (S9) yields a similar beta estimate ($\hat{\beta} = 0.88$) and alpha bias (0.32%), with the deterministic timing contributing only a modest additional effect. By limiting the compounding duration, both scenarios restrict the geometric growth of idiosyncratic errors, reducing the frequency of degenerate alpha estimates and partially restoring the accuracy of the market beta. However, even the most favorable holding-period scenario (S8) still exhibits substantially more beta attenuation ($\hat{\beta} = 0.91$ vs. 0.98) and alpha bias (0.30% vs. 0.05%) than the Driessen et al. (2012) benchmark. Tightening the alpha upper bound $\bar{\alpha}$ dramatically affects results: at $\bar{\alpha} = 0.5\%$ (S6), 32% of draws hit the bound, and beta is close to the benchmark ($\hat{\beta} = 0.87$), but at the cost of artificially constraining the alpha search space. This sensitivity reveals a fundamental tension in the two-factor model: the alpha term absorbs residual nonlinearity that the linear SDF cannot capture, and bounding it arbitrarily distorts the beta estimate.

Overall, the large remaining discrepancy between our simulation results and those of Driessen et al. (2012), even after adopting their shifted lognormal errors and simulated market returns, suggests that the favorable small-sample properties reported in Driessen et al. (2012) may be overly optimistic when the alpha upper bound is set more liberally ($\bar{\alpha}$ = 2%/month in our default configuration vs. $\bar{\alpha}$ = 1%/month in Driessen et al. (2012)). Across all scenarios with $\sigma = 30\%$, the beta estimates range from 0.62 to 0.91, well below the Driessen et al. (2012) benchmark of 0.98, confirming the estimator's sensitivity to the interplay between idiosyncratic noise, compounding Horizon, and alpha bounds.

Table 9: Comparison to simulation (Driessen et al., 2012, Table 1)

	Market Beta (β)					Alpha (α, monthly %)					
Scenario	True	Mean	SD	25%	75%	True	Mean	SD	25%	75%	% Bound
(S1) $n = 20$, $\sigma = 20\%$, Normal	1.00	0.84	0.40	0.61	1.05	0.00	0.22	0.28	0.04	0.32	0%
(S2) $n = 50$, $\sigma = 20\%$, Normal	1.00	0.89	0.29	0.74	1.03	0.00	0.13	0.16	0.03	0.19	0%
(S3) $n = 50$, $\sigma = 30\%$, Normal	1.00	0.62	0.72	0.12	1.03	0.00	0.68	0.70	0.13	1.08	0%
(S4) $n = 50$, $\sigma = 30\%$, Shifted LN	1.00	0.71	0.72	0.23	1.12	0.00	0.57	0.67	0.07	0.91	0%
(S5) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT	1.00	0.66	0.79	0.11	1.06	0.00	0.71	0.79	0.03	1.34	0%
(S6) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, $\bar{\alpha} = 0.5\%$	1.00	0.87	0.69	0.41	1.24	0.00	0.48	0.50	0.06	1.00	32%
(S7) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Det. Timing	1.00	0.70	0.79	0.19	1.12	0.00	0.65	0.80	-0.01	1.20	0%
(S8) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5	1.00	0.91	0.54	0.64	1.19	0.00	0.30	0.52	-0.03	0.47	0%
(S9) $n = 50$, $\sigma = 30\%$, Shifted LN, Sim. MKT, Max. Hold 5, Det. Timing	1.00	0.88	0.54	0.61	1.16	0.00	0.32	0.51	-0.01	0.53	0%
Driessen et al. 2012, Table 1											
Benchmark Estimates ($n = 50$, $\sigma_{\text{monthly}} = 30\%$)	1.00	0.98	0.32	0.80	1.15	0.00	0.05	0.2	-0.06	0.15	n/a

Notes: This table reports the small-sample properties of the generalized estimator for Horizon 0 (i.e., treating funds as held to maturity), mimicking the structure of Table 1 in Driessen et al. (2012). We simulate 1,000 Monte Carlo draws. All scenarios use 20 vintage years (1986–2005) with 15 deals per fund. The base case (S1) uses $n = 20$ funds per vintage, $\sigma = 0.2$ monthly idiosyncratic volatility, and a normal error distribution; scenarios S2–S9 deviate from these defaults as indicated in the table. The simulation dimensions vary as follows: n = number of funds per vintage-year portfolio; σ = monthly idiosyncratic standard deviation; *Normal* = Gaussian error term (default); *Shifted LN* = shifted lognormal error term (cf. Driessen et al. (2012)); *Sim. MKT* = total market return drawn from a shifted lognormal (S&P 500 1980–2003 moments) with a constant risk-free rate of 4% annualized, instead of conditioning on the historical path; *Det. Timing* = deterministic (evenly spaced) deal investment dates instead of random; *Max. Hold 5* = maximum holding period of a deal limited to 5 years (default 10 years); $\bar{\alpha}$ = upper bound on the alpha parameter in the optimization (default $\bar{\alpha}$ = 2%/month). The *% Bound* column reports the fraction of draws where the alpha estimate reached the upper optimization bound $\bar{\alpha}$. The final row presents the Benchmark Estimates obtained directly from Table 1 in Driessen et al. (2012) for comparative purposes. Alpha estimates are monthly and presented in percentages.

B.2 Comparison to empirical results

We now compare our empirical results directly with those published by Driessen et al. (2012). A key distinction is the underlying private market dataset: we use the Preqin cash flow database, which is also employed by Korteweg and Nagel (2016) and Ang et al. (2018), whereas Driessen et al. (2012) rely on the Thomson Venture Economics (TVE) fund-level dataset, which suffers from well-documented data quality issues such as forward-filling of stale KPIs (Harris et al., 2014). (Driessen et al., 2012, Table 5) report results for Buyout (BO) and Venture Capital (VC) funds using both a two-factor model (S&P 500 MKT + α) and the Fama and French

(1993) four-factor model (MKT, SMB, HML, and α). In their two-factor specification, VC exhibits a market beta of 2.73 and a highly negative annualized alpha of -12.3%, while BO yields a market beta of 1.31 and an annualized alpha of -4.8%. These strongly negative alpha estimates can be at least partially attributed to the aforementioned data errors in TVE; indeed, the BO alpha coefficient is statistically insignificant once bootstrap standard errors are taken into account, whereas the VC alpha remains significant. In their four-factor specification, the factor loadings are broadly plausible: BO exhibits $\hat{\beta}_{\text{MKT}} = 0.72$, $\hat{\beta}_{\text{SMB}} = 0.50$, and $\hat{\beta}_{\text{HML}} = 0.19$, while VC shows $\hat{\beta}_{\text{MKT}} = 2.20$, $\hat{\beta}_{\text{SMB}} = 1.83$, and $\hat{\beta}_{\text{HML}} = -0.48$. However, Driessen et al. (2012) note that the four-factor model does not significantly improve upon the two-factor specification and that the SMB and HML loadings are largely insignificant.

Table 10 and Table 11 present our empirical results for the single-factor (MKT), two-factor (α, MKT), three-factor (MKT, SMB, HML), and four-factor (α, MKT, SMB, HML) Fama and French (1993) models for BO, VC, and PE at Horizon 0, using the conservatively restricted sample with a maximum vintage-year cutoff of 2010.

We begin with the single-factor MKT model to establish a reliable baseline. Under this specification, market betas are consistently positive and theoretically plausible: the cross-validation fold averages ($\hat{\beta}_{\text{MKT}}$) are 1.20 for BO, 1.74 for VC, and 1.27 for PE under equal weighting (EW). The CV standard deviations verify that these estimates are highly stable across data subsets (e.g., CV standard deviations of 0.11 for BO and 0.06 for PE under EW). This result provides solid empirical backing that private equity returns exhibit strong, persistent co-movement with the broader market.

However, when explicitly allowing for an abnormal performance α intercept alongside the market factor, the estimation immediately collapses. In the two-factor model (α, MKT), the alpha parameter effectively acts as a sink that absorbs random, fund-specific noise compounded over a fund's lifetime. Even when restricting the sample to mature vintages, private equity datasets contain relatively few funds per year. As a result, there are simply not enough cross-sectional observations to adequately average out this idiosyncratic noise. Consequently, the optimizer overwhelmingly assigns inflated, positive alpha premiums (often hitting or nearing the 2% monthly bound, e.g., $\hat{\alpha} = 0.020$ for VC under EW and 0.011 for BO under FW), while the market betas are artificially driven toward zero or even turn negative (e.g., $\hat{\beta}_{\text{MKT}} = -0.13$ for BO and -0.10 for PE under FW). While asymptotic standard errors correspondingly explode or suspiciously round to zero (denoting entirely flat or near-degenerate regions in the objective function space), the CV standard deviations reveal the actual parameter variation across data subsets. For instance, the cross-validation fold dispersion for the MKT estimate exceeds 0.5 for both BO and PE under FW, confirming that the collapsed point estimates are highly sensitive to the sample and untrustworthy.

This pathological behavior extends directly to the multi-factor models. When estimating the three-factor model (MKT, SMB, HML) *without* an alpha term, the estimates roughly mirror the MKT-only model (e.g., BO $\hat{\beta}_{\text{MKT}} = 1.34$ under EW), and the CV analysis shows reasonable out-of-sample stability, even though the asymptotic standard errors are vast due to multicollinearity. The SMB loading

is consistently positive for BO and PE, but predominantly negative or near zero for VC. Notably, this sign for VC generally *reverses* the pattern reported by Driessen et al. (2012), who find a strongly positive VC SMB loading ($\hat{\beta}_{\text{SMB}} = 1.83$). The discrepancy likely reflects differences in the underlying datasets (Preqin vs. TVE) and sample periods, underscoring the severe fragility of multi-factor estimates in private equity.

But once the alpha term is reintroduced in the four-factor model (α, MKT, SMB, HML), the framework fails for the same reasons as the two-factor model. Because the alpha term absorbs the bulk of the remaining variance, the other factor loadings swing erratically: the asymptotic MKT coefficient for BO drops to essentially zero (-0.008 under FW), while SMB and HML loadings reach extreme values. The pervasive asymptotic standard errors of exactly (0.000) in these models are a telltale numerical artifact of a near-singular Hessian matrix hitting the boundary constraints. The CV analysis exposes this reality: rather than being perfectly estimated with zero error, the parameters swing erratically across hold-out sets, with CV standard deviations for the factor loadings routinely exceeding 0.4, formally proving that the multi-factor models are entirely unreliable when alpha is included.

In conclusion, our empirical evaluation provides a strong, objective corroboration of the simulation-based warnings from Section 3.3: with the limited cross-section of only 20–40 vintage-year portfolios currently available in commercial private equity databases, any generalized SDF model attempting to simultaneously extract multiple pricing factors alongside an explicit alpha parameter is severely under-identified given the available cross-section. The inclusion of an unconstrained or loosely constrained abnormal performance term systematically destroys the structural integrity of the entire factor model by confounding signal with cross-sectional noise. Therefore, until substantially more granular and independent cross-sectional data becomes cleanly available, we conclude that the single-factor market model remains the most robust and estimable specification for private equity SDF performance evaluation in our implementations.

Table 10: Estimation Results: 1-, 2-, 3-, and 4-Factor Models for Buyout and Venture Capital

	Equal-Weighted (EW)				Value-Weighted (FW)			
Model	α	MKT	SMB	HML	α	MKT	SMB	HML
Buyout (BO)								
1-Factor (Asymp.)		1.229				1.157		
		(0.769)				(0.994)		
1-Factor (CV)		1.198				1.100		
		(0.110)				(0.160)		
2-Factor (Asymp.)	0.009	0.147			0.011	-0.129		
	(0.000)	(0.080)			(0.071)	(11.302)		
2-Factor (CV)	0.007	0.411			0.010	-0.128		
	(0.004)	(0.518)			(0.005)	(0.631)		
3-Factor (Asymp.)		1.340	1.330	0.533		1.221	2.025	0.439
		(1.420)	(4.754)	(1.105)		(15.321)	(51.481)	(5.521)
3-Factor (CV)		1.303	1.167	0.498		1.165	1.746	0.330
		(0.164)	(0.389)	(0.120)		(0.160)	(0.489)	(0.240)
4-Factor (Asymp.)	0.007	0.288	0.454	0.079	0.008	-0.008	1.255	-0.286
	(0.004)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.005	0.676	0.303	-0.004	0.008	0.276	0.787	-0.110
	(0.003)	(0.392)	(0.284)	(0.213)	(0.004)	(0.505)	(0.404)	(0.604)
Venture Capital (VC)								
1-Factor (Asymp.)		1.850				1.585		
		(1.743)				(1.600)		
1-Factor (CV)		1.743				1.584		
		(0.411)				(0.361)		
2-Factor (Asymp.)	0.020	0.442			-0.001	1.691		
	(0.079)	(13.131)			(0.004)	(0.804)		
2-Factor (CV)	0.013	0.772			0.004	1.387		
	(0.009)	(0.206)			(0.010)	(0.558)		
3-Factor (Asymp.)		1.985	-0.296	1.487		1.931	-0.407	1.410
		(0.710)	(1.765)	(0.289)		(0.931)	(1.723)	(0.419)
3-Factor (CV)		1.893	-0.303	1.253		1.871	0.062	1.102
		(0.456)	(0.429)	(0.903)		(0.645)	(0.731)	(1.412)
4-Factor (Asymp.)	0.004	1.587	-0.675	1.679	-0.002	1.843	-0.173	0.154
	(0.001)	(0.000)	(0.000)	(0.000)	(0.005)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.012	1.185	0.006	1.645	0.004	1.628	-0.103	1.188
	(0.009)	(0.385)	(0.516)	(1.763)	(0.009)	(0.541)	(0.709)	(1.966)

Note: This table presents the estimation results of the single-factor (MKT), 2-factor (α, MKT), Fama-French 3-factor (MKT, SMB, HML), and 4-factor (α, MKT, SMB, HML) models. Cash flows are discounted to horizon 0. Both asymptotic (Asymp.) full-sample estimates and cross-validation (CV) fold averages are reported. Fold dispersions (standard deviations of the estimates across CV folds) are reported in parentheses below the estimates. The estimation uses Equal-Weighted (EW) and Value-Weighted (FW) cash flows. The upper bound for α was constrained to 0.02.

Table 11: Estimation Results: 1-, 2-, 3-, and 4-Factor Models for Private Equity

	Equal-Weighted (EW)				Value-Weighted (FW)			
Model	α	MKT	SMB	HML	α	MKT	SMB	HML
Private Equity (PE)								
1-Factor (Asymp.)		1.274				1.130		
		(0.799)				(1.029)		
1-Factor (CV)		1.268				1.065		
		(0.062)				(0.170)		
2-Factor (Asymp.)	0.004	1.005			0.010	-0.102		
	(0.007)	(1.438)			(0.067)	(11.349)		
2-Factor (CV)	0.004	0.903			0.009	-0.120		
	(0.002)	(0.291)			(0.005)	(0.626)		
3-Factor (Asymp.)		1.331	0.871	0.580		1.186	1.874	0.352
		(1.503)	(1.744)	(0.762)		(521.924)	(1737.701)	(254.679)
3-Factor (CV)		1.285	0.610	0.613		1.123	1.605	0.278
		(0.153)	(0.385)	(0.161)		(0.162)	(0.443)	(0.206)
4-Factor (Asymp.)	0.004	1.005	0.004	0.008	0.002	1.182	0.487	0.578
	(0.004)	(0.000)	(0.000)	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)
4-Factor (CV)	0.005	0.853	0.030	0.028	0.005	0.501	0.791	0.144
	(0.003)	(0.339)	(0.113)	(0.171)	(0.003)	(0.390)	(0.552)	(0.573)

Note: This table presents the estimation results of the single-factor (MKT), 2-factor (α, MKT), Fama-French 3-factor (MKT, SMB, HML), and 4-factor (α, MKT, SMB, HML) models. Cash flows are discounted to horizon 0. Both asymptotic (Asymp.) full-sample estimates and cross-validation (CV) fold averages are reported. Fold dispersions (standard deviations of the estimates across CV folds) are reported in parentheses below the estimates. The estimation uses Equal-Weighted (EW) and Value-Weighted (FW) cash flows. The upper bound for α was constrained to 0.02.